

Adroit Technologies

Protocol Driver

Name	Mitsubishi M2M
DLL	MIT_M2M



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1. Overview

The system allows telemetry functionality for efficient communication to highly distributed systems.

We have attempted to support the most important aspects required for a distributed system, namely:

- The need for time-stamped data so that events, rather than just register values, can be stored.
- An event is a register value (16-bit or 32-bit) and a timestamp with second resolution. Time of arrival (100ms separation) is attempted to be maintained if change events are registered faster than a second.
- The need to store in a buffer the events for later retrieval to a SCADA. This is especially needed at a time when communication becomes available and/or when the need arises to drill down and retrieve most specific data that would not normally require operator attention. For example, a temperature reading trend would only be of interest if say a high temp alarm were to be received.
- Efficient means of communication especially if data costs are an issue.

1.1. Supported Hardware

Adroit supports communication with the following PLCs:

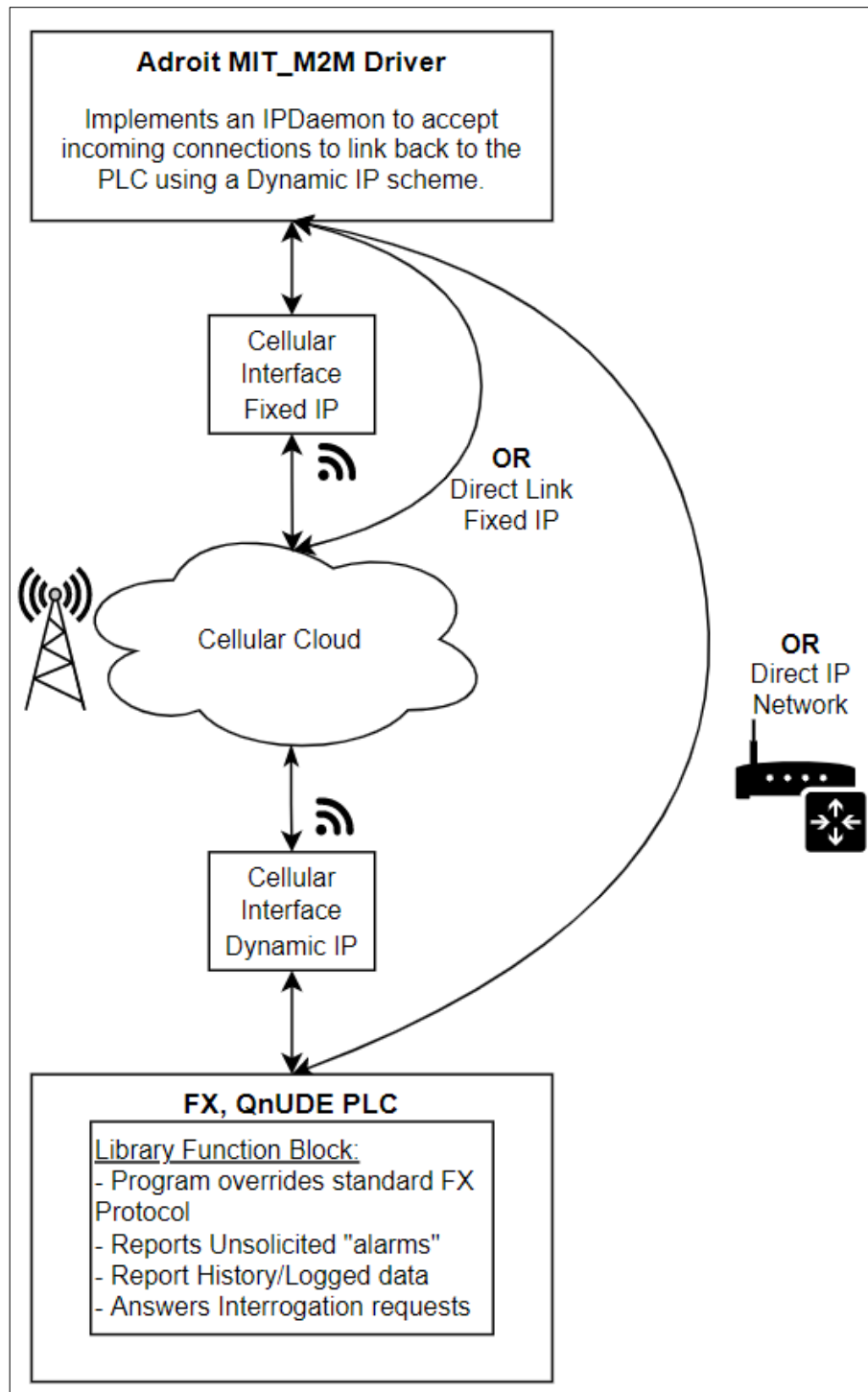
Mitsubishi FX3U with FX3U-ENET using the following "FXM2M613a" Library function block.

Mitsubishi FX5 PLC using the FX5U_M2M GX Works3 "FX5U_M2M634" program.

[As of 17/Feb/2022 found in FX5U_M2M_Rev634.gx3]

Mitsubishi QnUDE PLC using the following "QUDEM2M621" Library function block.

Mitsubishi L-Series PLC using the following "QUDEM2M621" Library function block.



2. Wiring

Standard Ethernet.

3. Device Configuration

3.1. Features and Operation:

The system supports the following "Areas" or data types scannable into Adroit:

- FX5 DATA Area: 500 data 16-bit WORDs or 250 32-bit WORDs can be mapped or any combination making up a total 500 WORDs. This is enforced because of the maximum packet size that can be transmitted in one transaction. (ONLY D registers can be monitored)
- Q/L DATA Area: 232 data 16-bit WORDs or 116 32-bit WORDs can be mapped or any combination making up a total 232 WORDs. This is enforced because of the maximum packet size that can be transmitted in one transaction. (ONLY D registers can be monitored) (500 WORD support still to come)
- CONFIG Area: 30 configuration WORDs.
- STATISTICS Area: 6 data usage DWORDs and 7 remote side statistical WORDs.
- STATUS Area: 3 status 16-bit WORDs, status bits, count WORD seconds to next poll, other status bits.
- ACTION Area: 2 control 16-bit packed WORDs.
- Data area is time-stamped (seconds resolution) data. Devices request time synchronization periodically (default is once per day, but this is configurable) but the master can set time at any point if required. (Time set is instantaneously, not gradually.)
- Manual poll instructions for all Areas, and a periodic DATA Area poll.
- Write commands for all Areas except STATISTICS.
- Unsolicited support for priority events of DATA Area.
- Unsolicited support for historical events of DATA Area.
- Logical addresses can be allocated to Priority (Alarmed) or Historical modes, with per logical address value deadbanding to reduce risk of chatter.
- The driver implements a daemon that will listen for and associate a channel to device link to allow incoming and outgoing data to a known device. The PLC is in control of the connection attempt and is expected to contact the master periodically via a LINK which the master will ACK. This allows for support for fixed and, part of the main design philosophy, dynamic non-public IPs typically found on public GPRS networks.
- FX Systems Historical buffer supporting 1000 (16-bit) events or 500 (32-bit) events.
- Q/L Systems Historical buffer supporting 7 500 (16-bit) events or 3750 (32-bit) events.
- FX Systems Priority buffer supports 250 (16-bit) events or 125(32-bit) events.
- Q/L Systems Priority buffer supports 250 (16-bit) events or 125(32-bit) events.

- Historical mode can be enabled/disabled. Automatic retrieval of history can be enabled/disabled. Historical buffer can be configured to be kept below a defined threshold.
- Any device may be put into reduced communications mode or 'Economy Mode' on a per device level, and/or per driver level, to allow for reduced comms load during high/peak periods. History buffer threshold exceeded reads and periodic interrogates are stopped during this time.
- Failover protection in case an invalid IP/port is configured the PLC will eventually timeout and go back to the last known "good" configuration (LKG) to reduce the risk of having to manually reprogram the configuration.
- A standard default configuration is also implemented for first boot-up and connection.
- A forced and fixed latency (quiet delay) between consecutive message transmissions from the master to the same PLC device is currently hardcoded to 500ms. Retry attempts are also hardcoded at 250ms.
- The following config data will result in dynamic change and persisted to keep the change valid between agent server restarts:
 - Enable priority events.
 - Enable history logging.
 - Keep buffer below %.
 - The actual %. – (this will only accept 25%, 50%, 75% and will automatically select the closest valid value.)
 - Periodic Update interval.

Local daemon configuration, however, will remain offline. Changes via Adroit's setup will require an Agent Server restart. Please note that even though station address is configurable via the CONFIG Area and can even be changed, this is specifically not persisted. The user is forced to manually allocate a new device configuration via setup.

Unknown contacts can thus come in via their default address assuming one at a time come alive, then a user can force a new address into the PLC via that same device. The PLC will then swing over to the newly provided device, thus freeing up the "unknown device" to wait for the next contact and the procedure can be repeated.

If a device does not answer, the PLC will failover back the original address allowing a retry.



WARNING

Port changes result in an entirely new Daemon being instantiated so devices cannot cross between communications Daemon.

- STATISTICAL data is available in the statistics area where data such as current daily and the last month's usage is available (this data is persisted and available between Agent Server restarts). Other PLC side data such as the number of times the modem has been reset, open errors, transactions etc. are also available. This must be manually polled via an ACTION bit.
- ACTION and STATUS WORDs are available for information on what modes the M2M is currently in, and for control of these functions including manual initiation of functions by toggling these bits as required.

- The M2M system can be put into "LIVE" mode, where it will send all DATA every 10 seconds. If not already on, it will set the "Auto history" to on which will attempt to flush the history buffer as well.
- Out of contact alert bit available.
- Start-up activity, the driver will time-synchronize the PLC if flagged to do so. A start-up data poll (interrogate) is sent, followed by sending a start-up config poll, then sending a start-up statistics poll. Enable unsolicited and historical modes if configured.
- Offline configuration is refreshed whenever the device is restarted, except that on the PLC side there are several latched regions in the file register that will persist between PLC power loss and program rewrites.
- As of rev620 for the Q-series, the driver has been implemented with a Primary/Secondary connection feature for back-up connections to an Adroit Master and Standby Server. In prior releases (for Q series, rev619 and earlier) this was referred to as Clustering. This has now been replaced by a more customisable dual-port feature mainly used for connection failures, allowing the system to connect an alternate Agent Server. The system can be altered to operate in Standalone or Dual-Port configuration through the CONFIG Area - CW28.

Refer to [Dual-Port and Standalone Mode](#) and [Appendix – Standalone and Dual-Port Details](#) in this document for more information on how this feature operates.

- As of rev35 of the Adroit driver, the M2M connection works with Adroit clustering both in Parallel and Shadow Scanning configurations with the new dual-port features.

3.2. PLC Performance Considerations: FX5

This system was designed for High Site monitoring where the number of I/O points is low. The system has limited buffer space for events so having large I/O DataMaps will cause it to easily become overcrowded. Attempting to increase these buffers will make the cycle time of the PLC much longer.

The Q Series PLC's are a lot faster, with typical cycle times in the low 10's of milliseconds so it is suggested to use a Q PLC when trying to use larger I/O maps, **and** the Q function block allows for larger buffers for event data, ~7500 vs ~1000 on the FX5.



PLEASE NOTE

The Q function block still only supports 232 currently.

As of Rev 30b for FX Systems the PLC can support a DataMap size of 500 16-bit (250 32-bit) registers.

Please consider the following cycle times (on a relatively idle PLC) when extending the DataMap:

DataMap size	Cycle time (Average)	Cycle Time (Max)
20 - (WORD signed)(0..19,0..3)	~17 ms	28,654 ms
50 - (WORD signed)(0..49,0..3)	~36 ms	47,698 ms

100	- (WORD signed)(0..99,0..3)	~68 ms	68,679 ms
200	- (WORD signed)(0..199,0..3)	~129 ms	137,917 ms
500 max	- (WORD signed)(0..499,0..3)	~315 ms	323,481 ms

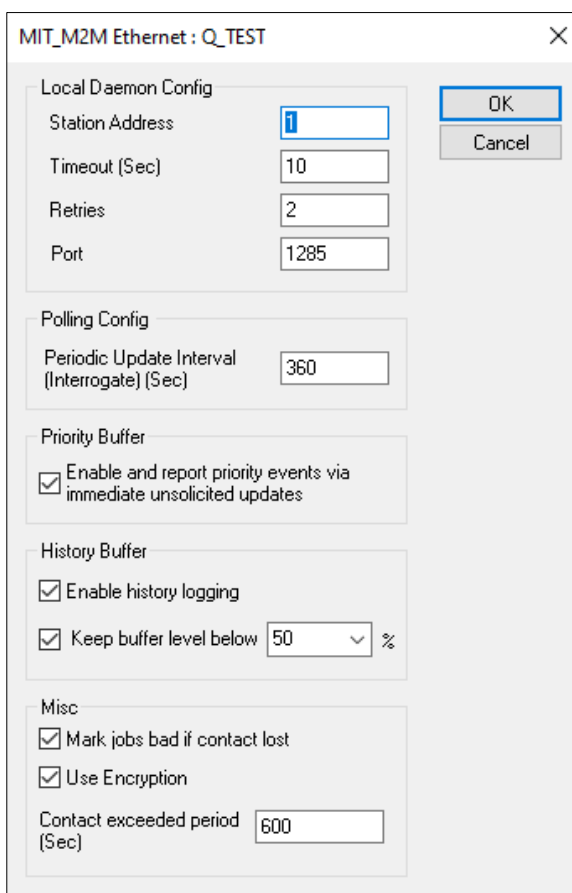


PLEASE NOTE

These cycle times do not include additional PLC logic that controls the values stored in the data registers for the DataMap. Additional logic on the PLC will also increase the cycle time and these values are only to indicate a baseline expected performance.

It is up to the user to determine an adequate cycle time for the PLC to perform all its functions.

3.3. Configuration Dialogs



The dialog box is titled "MIT_M2M Ethernet : Q_TEST". It contains several sections for configuration:

- Local Daemon Config:** Includes fields for Station Address (1), Timeout (Sec) (10), Retries (2), and Port (1285).
- Polling Config:** Includes a field for Periodic Update Interval (Interrogate) (Sec) (360).
- Priority Buffer:** Includes a checkbox for "Enable and report priority events via immediate unsolicited updates" which is checked.
- History Buffer:** Includes checkboxes for "Enable history logging" and "Keep buffer level below" (50 %), both of which are checked.
- Misc:** Includes checkboxes for "Mark jobs bad if contact lost" and "Use Encryption", both of which are checked. It also includes a field for "Contact exceeded period (Sec)" (600).

Buttons for "OK" and "Cancel" are located in the top right corner.

3.3.1. Local Daemon Config

Station Address – 16-bit WORD (1 – 65535). 0 is used as a special case for future use and is not to be used.

Timeout (seconds) – The number of seconds to wait on the protocol application layer (not the PLC).

Retries – The number of attempts not including the first attempt to retry a protocol function.

Port – A daemon will listen on this port.

3.3.2. Polling Config

Periodic update interval (Interrogate) (Seconds) – the period in seconds between each automatic Interrogation poll, which is a static snapshot at the current PLC time of the whole logical DATA area. [0 – disables]

3.3.3. Priority Buffer

Enable and report priority events via immediate unsolicited updates – (Default = Enabled) Will enable the priority alarming sent immediately via unsolicited update. The buffered events will be transmitted as soon as possible.

3.3.4. History Buffer

Enable History logging – (Default = Enabled) Will enable the logging of events that do not constitute immediate alarming for later retrieval via a manual poll for history, or automatic poll history every minute for 1 hour, or via an automatic threshold poll.



PLEASE NOTE

Regarding Auto reporting: If the buffer is not empty it will send 1 page of 500 historical events every 1 minute. This will only last for 1 hour and then the PLC will automatically reset it to off.

Keep buffer below... [checkbox] – (Default = Enabled) If enabled, the driver will wait for the history buffer to flag the appropriate level before attempting to completely flush it back to the master.

Keep buffer below... [level combo] - (Default = 50%) the level to be considered before flushing.

3.3.5. Miscellaneous

Mark Jobs bad if contact lost (Default = Enabled) The driver will mark all jobs bad when the out of contact bit is raised, when the device re-establishes contact the start-up polls are set to be polled as soon as possible.

4. Supported Addresses

PLC data address types supported by the FX_M2M Adroit Protocol driver:

4.1. Q/L-Series rev620, FX5 rev634 and Later:

Area	Min Range	Max Range	PLC Data Type	Boolean	Integer	Real	String
Data	0	Max D-Reg	WORD	-	DW0	DW0	-
			Integer	-	DI0	DI0	-
			DWORD	-	DD0	DD0	-
			LONG	-	DL0	DL0	-
			Float (single precision)	-	-	DF0	-
Config	0	29	WORD	-	CW0	CW0	-
			Integer	-	CI0	CI0	-
Statistics	0	6	WORD	-	EW0	EW0	-
	7	12	DWORD	-	ED7	ED7	-
Status	0	1	16-bit packed WORD	-	SW0	-	-
Action WORD	0	1	16-bit packed WORD	-	AW0	-	-

Note 1: Status and Action WORDs will be forced to WORD as this a 16-bit packed type, except SW1 which is a 16-bit counter.

Note 2: Statistics WORDs offsets from 0-6 are forced to WORD, 7-12 are forced to DWORD.

Note 3: Config areas can be scanned as DWORD or Long but the data is only 16 bits so they will just be formatted into the required type.

Note 4: Data area addressing refers to the physical 16-bit WORD location, 2 consecutive 16-bit locations can be mapped to one 32-bit value. DXy, where X refers to the final type that the value will be stored as in Adroit, y is the physical location in the memory buffer starting from 0. 32-bit values must be stored with the Least Significant WORD (LSW) in the first position and the Most Significant WORD (MSW) in the last position.

4.2. For revs earlier than rev620 and rev634:

Area	Min Range	Max Range	PLC Data Type	Boolean	Integer	Real	String
Data	0	Max D-Reg	WORD	-	DW0	DW0	-
			Integer	-	DI0	DI0	-
			DWORD	-	DD0	DD0	-
			LONG	-	DL0	DL0	-
			Float (single precision)	-	-	DF0	-
Config	0	15	WORD	-	CW0	CW0	-
			Integer	-	CI0	CI0	-
Statistics	0	6	WORD	-	EW0	EW0	-
	7	12	DWORD	-	ED7	ED7	-
Status	0	1	16-bit packed WORD	-	SW0	-	-
Action WORD	0	1	16-bit packed WORD	-	AW0	-	-

Note 1: Status and Action WORDs will be forced to WORD as this a 16-bit packed type, except SW1 which is a 16-bit counter.

Note 2: Statistics WORDs offsets from 0-6 are forced to WORD, 7-12 are forced to DWORD.

Note 3: Config areas can be scanned as DWORD or Long but the data is only 16 bits so they will just be formatted into the required type.

Note 4: Data area addressing refers to the physical 16-bit WORD location, 2 consecutive 16-bit locations can be mapped to one 32-bit value. DXy, where X refers to the final type that the value will be stored as in Adroit, y is the physical location in the memory buffer starting from 0. 32-bit values must be stored with the Least Significant WORD (LSW) in the first position and the Most Significant WORD (MSW) in the last position.

4.3. Addressing Details DATA AREA

4.3.1. Data area [DW0 – DW(max d-reg)] (READ and WRITEABLE)

The data area is the actual address of the D-Reg as defined in the map in the PLC.

The items defined do NOT need to be consecutive and it is entirely possible to have mixed memory 16-bit and 32-bit items randomly distributed across the map if the SCADA addressing matches.

From the PLC:

- If the PLC has defined an address size incorrectly as defined in the SCADA, the system will still operate normally except it will raise a status SW0, bit 11 MAP_MISMATCHED until the discrepancy is resolved.
- If say the PLC sends a DWORD and the SCADA expects a WORD, the SCADA is overridden to reflect a DWORD instead, as the compatible agents can accept a DWORD. Writes from the SCADA will however be cast to the lower WORD so the mismatch should be corrected as soon as possible.
- If say the PLC sends a WORD and the SCADA expects a DWORD, the SCADA accepts the data as is. Writes from the SCADA will however be cast to the lower WORD in the PLC so the mismatch should be corrected as soon as possible.



PLEASE NOTE

Consider the 3rd value in the first stack (DD2:2) - it is a DWORD, and it is logically the third item in the stack. It affects the physical address position of the next data point (DW4:3). This fourth data points to physical location that starts one value lower because of the DWORD item above it. This is very important as an INTERROGATE request will retrieve this whole data area, and if the scanning is not defined correctly, it will not unpack correctly.

4.4. Addressing Details CONFIG AREA

4.4.1. Config Area [CW0 – CW15] (READ and WRITEABLE (on some))



PLEASE NOTE

As of rev620 for the Q-series, the CONFIG Area has been expanded for CW0 – CW29, as well as bit 14 of ACTION Word 0 has been repurposed. These changes are currently not available for the FX PLCs. Please contact Adroit for further queries.

The configuration area is designed to allow the SCADA to see and possibly tweak a devices configuration. Use this with caution as an invalid or unreasonable parameter may inadvertently force the PLC into a state that can seriously limit communications or perhaps even break it.

- This area does get a start-up poll so that the device will be marked healthy.
- Manual read on Action WORD 2 (AW0), Bit 1 (2nd Bit). (Dig01 if marshal)
- Writes are sent via the tag change in Adroit while 'Output Enabled' is set to true. It is only accepted if it has been specified to not be read-only in the driver.

WORD 0	ID (Immediate)
WORD 1	IP Most Significant WORD (MSW) (Read-Only)
WORD 2	IP Least Significant WORD (LSW) (Read-Only)
WORD 3	Port (Read-Only)
WORD 4	Timeout (Immediate 100ms) [≥ 20] seconds
WORD 5	Retries (Immediate)
WORD 6	Live duration (Immediate 100ms)
WORD 7	Live reporting interval (Immediate 100ms) [≥ 100]
WORD 8	Auto History duration (Immediate / hours)
WORD 9	Auto History reporting Interval (Immediate 100ms) [≥ 100]
WORD 10	Real time clock request update interval (Immediate hours)
WORD 11	Links interval (immediate 100ms) [≥ 100]
WORD 12	M2M Software version (Read-Only)
WORD 13	<u>On Standalone:</u> NEW IP (MSW) (AW0 Reset Required)
	<u>On Dual-port:</u> (Available Only for Q-series Rev620, FX5 Rev634 and later)
	NEW IP on A (MSW) (AW14 Force Switch over required)
	<u>On Dual-port:</u> (Available Only for Q-series Rev620, FX5 Rev634 and later)
WORD 14	NEW IP on A (MSW) (AW14 Force Switch over required)
	<u>On Standalone:</u> NEW IP (MSW) (AW0 Reset Required)
	<u>On Dual-port:</u> (Available Only for Q-series Rev620, FX5 Rev634 and later)
	NEW IP on A (MSW) (AW14 Force Switch over required)
WORD 15	<u>On Standalone:</u> NEW IP (MSW) (AW0 Reset Required)
	<u>On Dual-port:</u> (Available Only for Q-series Rev620, FX5 Rev634 and later)
	NEW IP on A (MSW) (AW14 Force Switch over required)
	<u>On Dual-port:</u> (Available Only for Q-series Rev620, FX5 Rev634 and later)
WORD 16	NEW IP on A (MSW) (AW14 Force Switch over required)
	Keep Buffer level below (Immediate) [Internal to Adroit]
WORD 17	Periodic update interval (Immediate seconds) [Internal to Adroit]
WORD 18	Enable auto buffer levelling (immediate 0 or >0) [Internal to Adroit]

Available Only for Q-series Rev620, FX5 Rev634 and later:

WORD 19	On Standalone: NEW IP on B (MSW) (Immediate) [Caution: these changes will not be fully implemented until you change to Dual-port mode, these will still be stored in the config words, but no flags will be set indicating so when on Standalone.] On Dual-port: NEW IP on B (MSW) (AW14 Force Switch over required)
WORD 20	NEW IP on B (LSW) (Immediate) NEW IP on B (LSW) (AW14 Force Switch over required)
WORD 21	On Standalone: NEW Port on B (Immediate) On Dual-port: NEW Port on B (LSW) (AW14 Force Switch over required)
WORD 22	Last Known Good IP on A (MSW) (Read-Only)
WORD 23	Last Known Good IP on A (LSW) (Read-Only)
WORD 24	Last Known Good Port on A (Read-Only)
WORD 25	Last Known Good IP on B (MSW) (Read-Only)
WORD 26	Last Known Good IP on B (LSW) (Read-Only)
WORD 27	Last Known Good Port on B (Read-Only)
WORD 28	Standalone/Dual-Port Mode (Immediate, 0=Standalone, >0 =Dual-Port)
WORD 29	Standalone/Dual-Port flags. This is a bit packed WORD: Bit 0 = 1: Currently Connected on A, Bit 0 = 0: Currently Connected on B Bit 1 = 1: Reset_M2M Required Bit 2 = 1: ForceSwitch Required

Considerations for new IP to set it to say 10.10.10.21 port 1234,

1010(hex) = 4112(dec) => 13: NEW IP MSW
1021(hex) = 4129(dec) => 14: NEW IP LSW
1234(hex) = 4660(dec) => 15: NEW PORT

Then, depending on if the system is on Standalone configuration, a Reset_M2M triggered by Action WORD AW0, Bit 1 would be required.

(Available Only for Q-series Rev620, FX5 Rev634 and later):

As part of these changes, the user can now make connection changes to both the primary (A) and the secondary (B) through CW13, CW14, CW15 and CW19, CW20, CW21 respectively. When a new change is made to one of the connection details, the system will flag the pending required action in parts of CW29. The mechanisms operate as follows:

- When on Standalone, any changes to the primary connection require a Reset_M2M (AW0, Bit 1) and CW29 Bit 1 will be set to indicate a pending required reset.



PLEASE NOTE

This flag will only clear when a new change is brought in, and a successful connection has been made to it. Refer to ['Change Connection Details \[Maintaining the current connection\]: Standalone'](#) in the Appendix for more information.

- When on Standalone, any changes to the secondary connection are brought do not affect the connection details. CW29 will remain unaffected, and no flags will be set. However, making changes to these connection details and then changing to dual-port mode will then bring in the changes and set the necessary flags. Refer to 'Change Connection Details [Updating Secondary (B)]: Standalone' in the Appendix for more information.
- When on Dual-port, any changes to the primary connection or secondary will flag ForceSwitch (AW0, Bit 14) and CW29 Bit 2 will be set to indicate a pending required switch. However, the operator has two choices and does not necessarily have to switch the connection. The choices are summarized as follows:
 - A change on the current connection has been sent by the operator but the operator wants the connection to re-establish on the same current connection (in terms of primary and secondary), then the operator can perform trigger a Reset_M2M. The new connection details will be loaded but connection will first try to connect on the same dual-port connection. For example, the operator is currently on the secondary connection and makes a change on the secondary, but the operator wishes to stay on the secondary, then they can perform a Reset_M2M action. Refer to 'Change Connection Details [Maintaining the current connection]: Dual-Port' in the Appendix for reference.
 - A change on the alternative connection details has been sent by the operator but the operator wants the connection to re-establish on the alternative connection (in terms of primary and secondary), then the operator can perform trigger a ForceSwitch. The new connection details will be loaded, and the connection will then first try to connect on the alternative dual-port connection. For example, the operator is currently on the secondary connection and makes a change on the primary, but the operator wishes to change to the primary to verify a valid connection, then they can perform a ForceSwitch action. Refer to '[Change Connection Details \[Applied to alternate connection\]: Dual-Port](#)' in the Appendix for reference.



PLEASE NOTE

The flags specified in Bit 1 and 2 of CW29 will only clear when a new change is brought in, and a successful connection has been made to it. For example, changing the details on the primary connection while connected on the secondary and performing a Reset_M2M action will not clear that ForceSwitch flag, since an attempt on the new connection was not attempted. This is the same if the new connection details do not result in a valid connection. Refer to '[Change Connection Details \[Maintaining the current connection\]: Dual-Port](#)' and '[Change Connection Details \[Applied to the alternate connection\]: Dual-Port](#)' in the Appendix for more information.



WARNING

Adroit recommends that changes on both A and B simultaneously be avoided where possible. Both connections have their respective last known good connections that will be loaded in the event of connection failures; however, causing changes to these details while deactivating all the last known good connections could lead to a situation where the system may not connect and cannot restore any previous connection – therefore breaking communications. In this scenario, a manual reconfiguration will be required for the affected PLC. It is recommended that one change is made at a time, then verifying that change and finally ensuring a stable connection before implementing the next change.



PLEASE NOTE

ID is used to determine what comms resource will feed it. A write into this without an appropriate comms resource available will cause the system to not be able to communicate, and a failover to last known good config will result.

Values in brackets (e.g. [≥ 20]) mean the PLC will enforce a minimum or maximum as appropriate and ignore any write outside of these bounds.

Immediate means no reset is required.

Some values are denoted in various time types, such as the value in this address must be multiplied with that type. For example, 230 in CW7 = $230 \times 100\text{ms} = 23 \text{ Seconds}$.

CW29 does not update nearly as often as the Status WORD 0 and 1. This means that alarming on this bit packed WORD should be avoided. The operator can force it to update by triggering the READ CONFIG action (AW1, Bit 1) but doing this regularly will increase the total bytes sent and therefore data costs. It is recommended that after any changes that require an M2M reset, or force switch should be performed soon after the connection details have been changed.

4.5. Addressing Details STATISTICS AREA

4.5.1. Statistics Area [EW0 – ED12] (READ only)

Statistics area is meant to give a user some audit info regarding communications data usage etc.

- This area does get a start-up poll so that the device will be marked healthy.
- Manual read on Action WORD 1 [AW1], 5th bit. (Dig04 if marshal)
- Writes are sent via the tag change in Adroit, if output enabled, and accepted if not read-only.
- The PLC statistics EW0-7 can be reset by using the AW0 8th bit. The data Audits ED7-12 cannot be reset.

EWX where X is between 0 and 6, and EDX where X is between 7 and 12:

WORD 0	Number of reboots (16-bit)
WORD 1	Number of retries (16-bit)
WORD 2	Number of messages sent (16-bit)
WORD 3	Number of messages received (16-bit)

WORD 4	Number of open errors (16-bit)
WORD 5	Number of send errors (16-bit)
WORD 6	Number of receive errors (16-bit)
WORD 7	Last month's bytes TX (32-bit)
WORD 8	Last month's bytes RX (32-bit)
WORD 9	Current month's bytes TX (32-bit)
WORD 10	Current month's bytes RX (32-bit)
WORD 11	Today bytes TX (internal)
WORD 12	Today bytes RX (internal)

4.6. Addressing Details STATUS AREA

4.6.1. Status Area [SW0] (Read-Only internal returned automatically)

Since every response from the PLC contains the current status, the status area WORD is updated from any transaction returned by the PLC.

This is the only AREA that will not be marked bad and grey out, so that some bits may be used during offline periods for example "Contact exceeded"

It is recommended to scan to a marshal or multistate to allow for easy access to bits.

SW0 where each bit defined below:

BIT 0	STARTED - M2M engine is started in PLC: data acquisition and storage enabled
BIT 1	NEEDTIME - M2M engine needs time: PLC asking for time synchronization
BIT 2	LIVE - M2M is in Live mode and will push snapshots of data to master regularly*
BIT 3	PRIORITY_STARTED - Priority data functions are started in PLC
BIT 4	PRIORITY_OVER - Priority buffer has overflowed and is wrapping
BIT 5	HIST_STARTED - History buffer functions are running
BIT 6	HIST_OVER - History buffer has overflowed and is wrapping
BIT 7	HIST_AUTO - History will empty the history buffer to the master regularly*
BIT 8	LINKS_ACTIVE - Link messages / keep alive's are sent updates to master regularly*
BIT 9	CONTACT_EXCEEDED - PLC has not reported in for set expected time*
BIT 10	ECONOMY_MODE - PLC is in data economy mode, all automatic history updates and periodic interrogates will not work, including Live Mode. Excludes Manual Poll's or Priority Updates (Alarms/Events)
BIT 11	MAP_MISMATCHED - PLC and Adroit have mismatched tables. Ex If data map in PLC is defined as a Double-WORD and Adroit is set to scan as a WORD or visa-versa this flag will be raised.
BIT 12	HIST_MSB - if on history buffer size is increased by 50**
BIT 13	HIST_LSB - if on history buffer size is increased by 25**
BIT 14	PRIORITY_MSB - if on unsolicited buffer size is increased by 50**
BIT 15	PRIORITY_LSB - if on unsolicited buffer size is increased by 25**

* The period that the function activates is defined in the config area and has a default if not explicitly scanned.

** Looking at these 2 bits the state 00 = 0-25%, 01=25-50%, 10=50-75%, 11=75-99%

4.6.2. Status Area [SW1] (Read-Only internal returned automatically)

The count of the remaining seconds left before the next poll.

4.7. Addressing Details ACTION AREA

4.7.1. Action Area [AW0 – AW1] (READ internal pulsed will return to zero, WRITE)

The Action WORDs give the SCADA more control over the various functions of the M2M allowing for more precise manual control.

It is recommended to scan to a marshal or multistate to allow for easy access to bits.

This area is non-latching the driver will reset all bits back to zero. Multiple bits may be sent at the same time, but they will be executed serially and on first set first executed basis.

AW0 where each bit defined below:

BIT 0	START_M2M – Start the M2M engine
BIT 1	RESET_M2M - Clear all buffers and load new IP configuration
BIT 2	GO_LIVE – Request the M2M to send full updates more frequently*
BIT 3	ENABLE_PRIORITY – Enable priority buffer functions
BIT 4	ENABLE_HISTORY - Enable history buffer functions
BIT 5	ENABLE_AUTO_HISTORY – Start sending history updates for a period*, frequently*
BIT 6	ENABLE_LINKS – Enable links or keep-alive's (NB- contact may be infrequent if disabled)
BIT 7	RESET_STATISTICS – M2M statistics will be cleared (not data stats)
BIT 8	STOP_M2M – Stop M2M engine except comms functions
BIT 9	STOP_LIVE – Stop sending full updates more frequently.
BIT 10	DISABLE_PRIORITY - Disable priority buffer functions
BIT 11	DISABLE_HISTORY – Disable history buffer functions, no logging will take place.
BIT 12	DISABLE_AUTO_HISTORY - Stop sending history updates immediately
BIT 13	DISABLE_LINKS - Disable links or keep-alive's (NB- contact may be infrequent if disabled)
BIT 14	FORCESWITCH – (Available Only for Q-series Rev620, FX5 Rev634 and later) On Dual-Port mode, switch between Primary and Secondary stored connections.
BIT 15	RESET_OVERFLOW_FLAGS – Clears the overflow status bits

AW1 where each bit defined below:

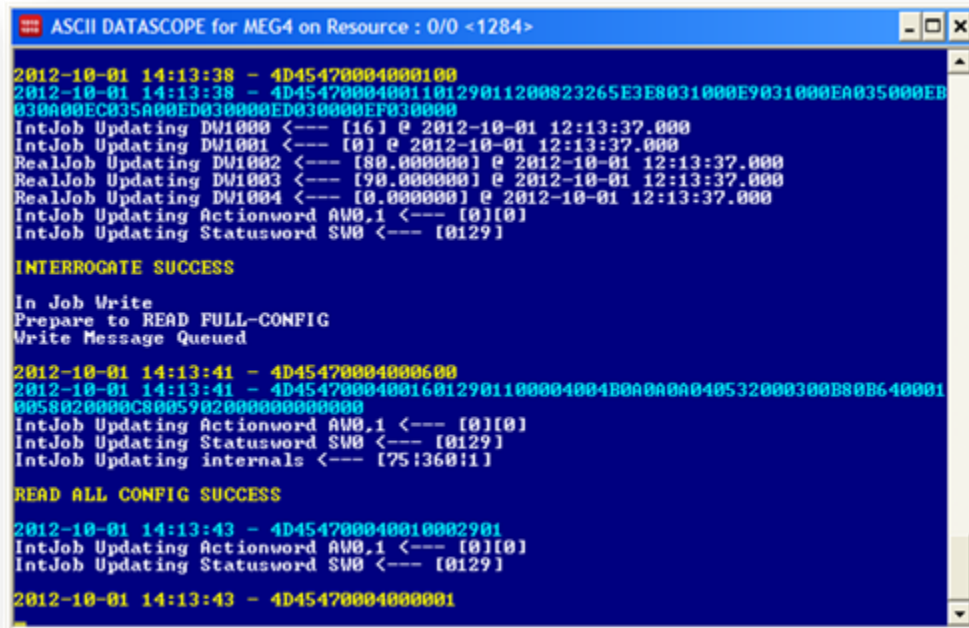
BIT 0	READ_DATA - Read full data area now (Interrogate)
BIT 1	READ_CONFIG - Read full config area now
BIT 2	SET_TIME - Write time to M2M now {ignores time required bit}
BIT 3	READ_HISTORY - Read entire history buffer now
BIT 4	READ_STATISTICS - Read full statistics area now
BIT 5	<i>(Reserved for future use)</i>
BIT 6	<i>(Reserved for future use)</i>
BIT 7	<i>(Reserved for future use)</i>
BIT 8	ENABLE_ECONOMY - Enable Economy mode for device only
BIT 9	ENABLE_ALL_ECONOMY - Enable Economy mode for all M2M devices
BIT 10	DISABLE_ECONOMY - Disable Economy mode for device only
BIT 11	DISABLE_ALL_ECONOMY - Disable Economy mode for all M2M devices
BIT 12	<i>(Reserved for future use)</i>
BIT 13	<i>(Reserved for future use)</i>
BIT 14	<i>(Reserved for future use)</i>
BIT 15	<i>(Reserved for future use)</i>

- * The period that the function activates is defined in the config area and has a default if not explicitly scanned.

5. Protocol Definition and Considerations

5.1. Datascope

Typical information available on the datascope.



```

ASCII DATASCOPE for MEG4 on Resource : 0/0 <1284>

2012-10-01 14:13:38 - 4D45470004000100
2012-10-01 14:13:38 - 4D4547000400110127011200023265E3E8031000E9031000EA035000EB
030A00EC035A00ED030000ED030000EF030000
IntJob Updating DW1000 <--- [16] @ 2012-10-01 12:13:37.000
IntJob Updating DW1001 <--- [0] @ 2012-10-01 12:13:37.000
RealJob Updating DW1002 <--- [80.000000] @ 2012-10-01 12:13:37.000
RealJob Updating DW1003 <--- [90.000000] @ 2012-10-01 12:13:37.000
RealJob Updating DW1004 <--- [0.000000] @ 2012-10-01 12:13:37.000
IntJob Updating Actionword AW0.1 <--- [0][0]
IntJob Updating Statusword SW0 <--- [0129]

INTERROGATE SUCCESS

In Job Write
Prepare to READ FULL-CONFIG
Write Message Queued

2012-10-01 14:13:41 - 4D45470004000600
2012-10-01 14:13:41 - 4D454700040016012701100004004B0A0A0A040532000300B80B640001
0058020000C800590200000000000000
IntJob Updating Actionword AW0.1 <--- [0][0]
IntJob Updating Statusword SW0 <--- [0129]
IntJob Updating internals <--- [75:360:1]

READ ALL CONFIG SUCCESS

2012-10-01 14:13:43 - 4D454700040010002701
IntJob Updating Actionword AW0.1 <--- [0][0]
IntJob Updating Statusword SW0 <--- [0129]

2012-10-01 14:13:43 - 4D45470004000001
  
```

5.2. Dual-Port and Standalone Mode

5.2.1. Description

From rev620 on the Q-series and FX5 rev634 onwards, the cluster mechanism has been adjusted to provide additional features. The purpose of this system is to provide an additional communication channel to communicate back to the Master via, should communications of the primary connection fail. For a full breakdown of how this system operates, refer to the Appendix.

Along with this, additional CONFIG WORDs have been added for additional system information as it relates to the dual-port configuration, such as additional writeable addresses for the secondary connection (B), the Last Known Good (LKG) connections on both the primary (A) and secondary connections, the ability to switch between Standalone and Dual-Port modes of operation as well as additional flags to indicate the state of the dual-port configuration stored in CW29. In this 16-bit WORD:

- Bit 0: indicates whether the connection is currently on A or B
- Bit 1: indicates that a Reset_M2M on AW0, Bit 1 is required as it has detected new connection address details (IP and/or Port) in a way that requires a reset. This will be raised/set only when a change is detected on the primary (A) CONFIG WORDs 13, 14 or 15 while in **Standalone** mode.
- Bit 2: indicates that a ForceSwitch on AW0, Bit 14 is required as it has detected new connection address details (IP and/or Port) in a way that requires a reset. This will be raised/set only when a change is detected on either CONFIG WORDs 13,14 or 15 and on 19, 20 or 21 respectively while in **Dual-Port** mode, however a Reset_M2M can be used should the user wish to remain on the current connection.



PLEASE NOTE

The dual-port and standalone configurations have some complexity in how they operate, so it is important to understand its operation before making use of the system along with the additionally included CONFIG WORDs. Please do refer to the Appendix for a full breakdown of its functioning.

Refer to [Appendix B – Standalone/Dual-Port Details](#) for a detailed description of this mechanism.

5.3. M2M Configuration with Adroit Cluster Server (Shadow Scanning vs Parallel Scanning)

As of mit_m2m.dll rev35, additional features have been implemented to work with Adroit clustering. Dual-port mode of the driver allows the PLC to communicate with standalone Agent Servers, as well as servers in a Cluster system architecture. However, there are differences in how the driver operates depending on how the Cluster system is configured in terms of I/O Options, which is described below.

5.3.1. Shadow Scanning

When dealing with Shadow Scanning, the PLC will only be able to communicate with a device that is configured as Master. The socket on which the device attempts to connect on will remain closed while that Agent server remains in Standby mode. Any attempt to switch the PLC connection from the server flagged as Master to the connection flagged as Standby will still cause the connection to close and the PLC to attempt to connect to the Standby, only to fail after the timeouts and retries have been exceeded before returning connection to the original Master Agent server. However, if the Agent Server that the PLC is connected on switches from Master to Standby, the PLC will receive a command to connect on the alternative connection.

5.3.2. Parallel Scanning

When dealing with Parallel Scanning, the PLC can communicate with either the **Master** or **Standby** Agent Server, though only through one of the connections at a time. Switching Agent Servers between Master and Standby will not cause the PLC to change connections unless it has failed to communicate to that Agent Server.

It is recommended that when using Parallel Scanning that both '**Enable writes when in standby mode**' and '**Enable reads when in standby mode**' be enabled.

5.4. Hardware Considerations

GPRS Routers tested:

- **Adroit Wireless router – eTung MR900**
- **MOXA**
- **Duotech Beres ETP (Ethernet Terminal Plus)**

5.5. PLC Considerations

5.5.1. PLC Library

The Adroit M2M driver is designed to be used in conjunction with a library for the Mitsubishi FX3U PLC with an E-Net module.

The library will facilitate communications by interpreting a map specified by the user and using the register locations and dead bands supplied in the map to build communications messages that can be understood by the SCADA.

5.5.2. Resource Requirements (VERY IMPORTANT)

The PLC library will require the following resources on the PLC to function properly:

Q PLC:

- 500 Devices to be auto-assigned by GX-Works2
- Program memory/Device Memory considerations:
 - 10 270 Steps/41 080 Bytes for the PLC program (Base example program with DataMap size of 20 WORDs)
 - Incrementing the value of DataMapLength (with associated BuildDataMap entry) increases program size by approximately 8 Steps/32 Bytes (not including additional PLC logic to control the value stored in the additional DataMap indices).
- V1 through V4 index variables.
- A fixed buffer of the size of the DataMap to be implemented (under GLOBAL_LABELS). This buffer needs to be four times the size of the number of datapoints to be monitored, at some D address.
- Reserved registers:
 - The first 8000 R Registers
 - File registers: ZR0 – ZR35526 (Type 2 Latched)
 - Data registers: starting from D5000 to D5000+4 × (DataMap fixed size)

FX5 PLC:

- Program memory/Device Memory considerations:
 - 40 662 Steps for the PLC program (Base example program with DataMap size of 20 WORDs)
 - Incrementing the value of DataMapLength (with associated BuildDataMap entry) increases program size by approximately 16 Steps (not including additional PLC logic to control the value stored in the additional DataMap indices).
- Data registers D0, D1 The DataMap address e.g. above D2000 through (D2000 + (DataMapSize × 4)) are reserved.
- R0 – R8026 are reserved. (R8020-R8026 Type 2 Latched)
- Z1-Z7 indexers are reserved.

Refer to [Appendix – Memory Requirements](#) for additional information.

5.5.3. Program Configuration

For the library to communicate it is essential that two programs be implemented in the PLC. The first of these programs must contain the initialisation of the DataMap and should be configured to run only once at start-up of the PLC with the highest priority. The second program will contain the implementation of the protocol and will run cyclically.

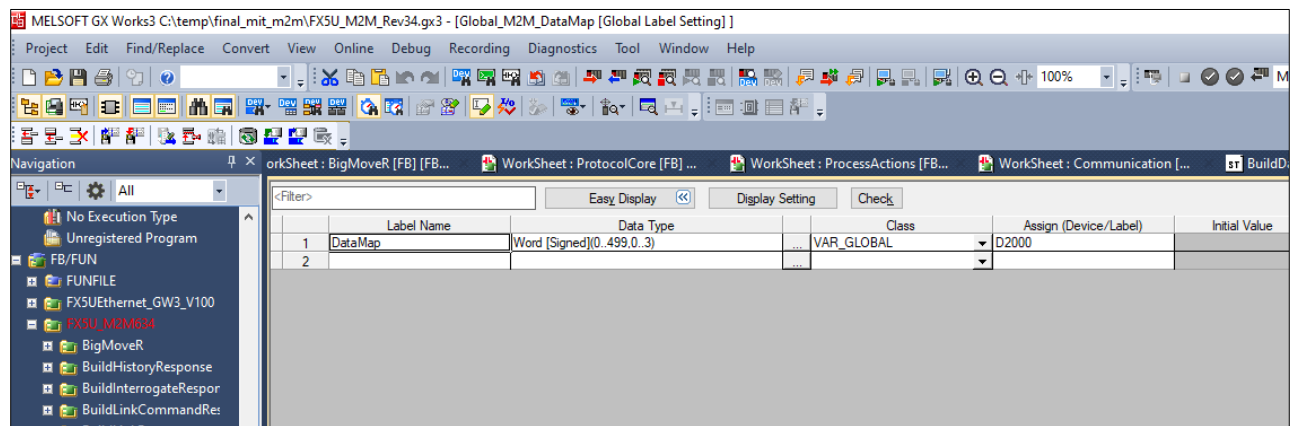
5.5.4. Initialise DataMap Program

It is recommended that the initialize DataMap program be written in structured text (ST). To implement this program a global variable must be created that contains a two-dimensional array. In this example it will be called DataMap.

GX Works 2: (QUDE PLC)

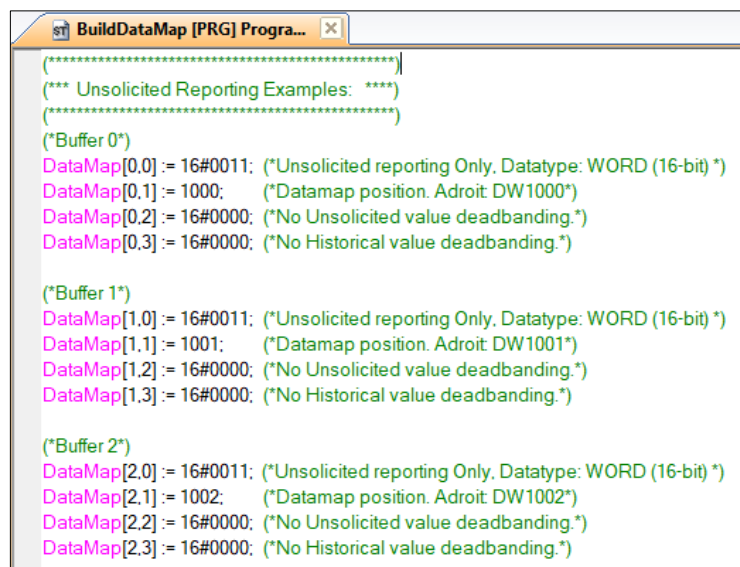
43	VAR_GLOBAL	DataMap	Word(Signed)(0..200,0..3)	D5000	%MW0.5000
----	------------	---------	---------------------------	-------	-----------

GX Works 3 (for FX5 PLC):

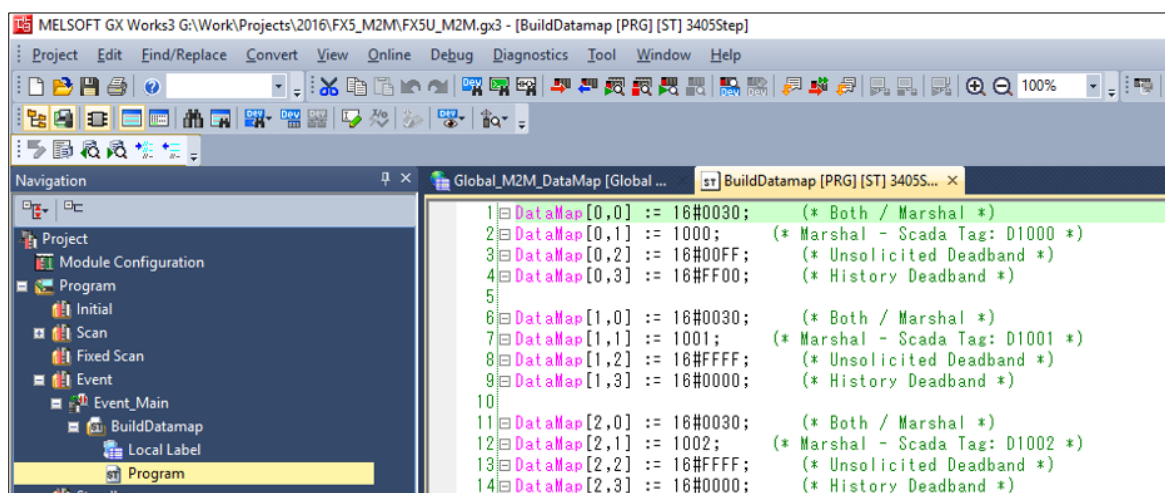


This global variable must have a fixed start address as this address will be used as a reference in the other program. The size of width of this array must be four elements and the maximum length of the array is the number of supported elements if 16-bit, and half if 32-bit or any combination thereof. The format of the program that builds the DataMap should be as follows:

GX Works 2: (QUDE PLC)



GX Works 3 (for FX5 PLC)



- The first column of the DataMap will contain a function WORD that tells the library how to treat the data.
- The second column shows the location of the data that must be communicated.
- The third column contains the deadband or bitmask that will be used for unsolicited processing.
- The fourth column contains the deadband or bitmask that will be used for historical processing.

The breakdown of the values will be as such:

DataMap[x,0]:

This is the function code that will tell the library how to process the value and can have the following values:

16#00YZ

Z signifies the datatype of the register that is to be monitored.

Z Value	DataType	Meaning
0	Marshal	A 16-bit WORD containing packed digital signals.
1	WORD	A 16-bit value.
9	DWORD	A 32-bit value with the LSW in the first position.

Y signifies whether the data is to be treated unsolicitedly or as historical values. For any datatype other than a marshal a setting of unsolicited will override a setting of historical.

Y Value	Meaning	Valid DataTypes
1	Data will be treated for Unsolicited reporting only.	All.
2	Data will be treated for Historical reporting only.	All.
3	Data will be treated for both Unsolicited and historical reporting	Marshals only.

DataMap[x,1]:

This contains the indexed position of the data to process. For example, if this value is 1234 then the processing will be applied to D1234 and if this value is 4567 then the processing will be applied to D4567.

DataMap[x,2]:

This value has two different meanings depending on which data type the row contains. If the row contains a Marshal, then this value will constitute a bitmask that is applied to the value to see if an unsolicited message is to be sent. If the row contains an analog datatype then this value constitutes the deadband that will be applied to the value for unsolicited reporting. This dead band is non-inclusive, therefore if a value of 2 is set here, then an unsolicited message will only be sent when the value changes by more than 2.

DataMap[x,3]:

This value has two different meanings depending on which data type the row contains. If the row contains a Marshal, then this value will constitute a bitmask that is applied to the value to see if an Historical message is to be logged. If the row contains an analog datatype then this value constitutes the deadband that will be applied to the value for historical reporting. This deadband is non-inclusive, therefore if a value of 2 is set here, then an unsolicited message will only be sent when the value changes by more than 2.

Examples:

```
DataMap[0,0] := 16#0021;
DataMap[0,1] := 1234;
DataMap[0,2] := 16#0000;
DataMap[0,3] := 16#0001;
```

This is a 16-bit WORD variable that is stored in D1234. This WORD is configured to not be reported unsolicitedly, but to be logged to the historical buffer until such time as the buffer is requested. A deadband value of 1 is applied, therefore the data will only be logged to the buffer if D1234 changes by two or more.

```
DataMap[1,0] := 16#0019;
DataMap[1,1] := 1348;
DataMap[1,2] := 16#000A;
DataMap[1,3] := 16#0000;
```

This is a 32-bit DWORD variable that is stored in D1348 and D1349 with the Least Significant WORD being stored in D1348. This DWORD is configured to be reported unsolicitedly, but not to be logged to the historical buffer until such time as the buffer is requested. A deadband value of 10 is applied, therefore the data will only be sent if the DWORD changes by 11 or more.

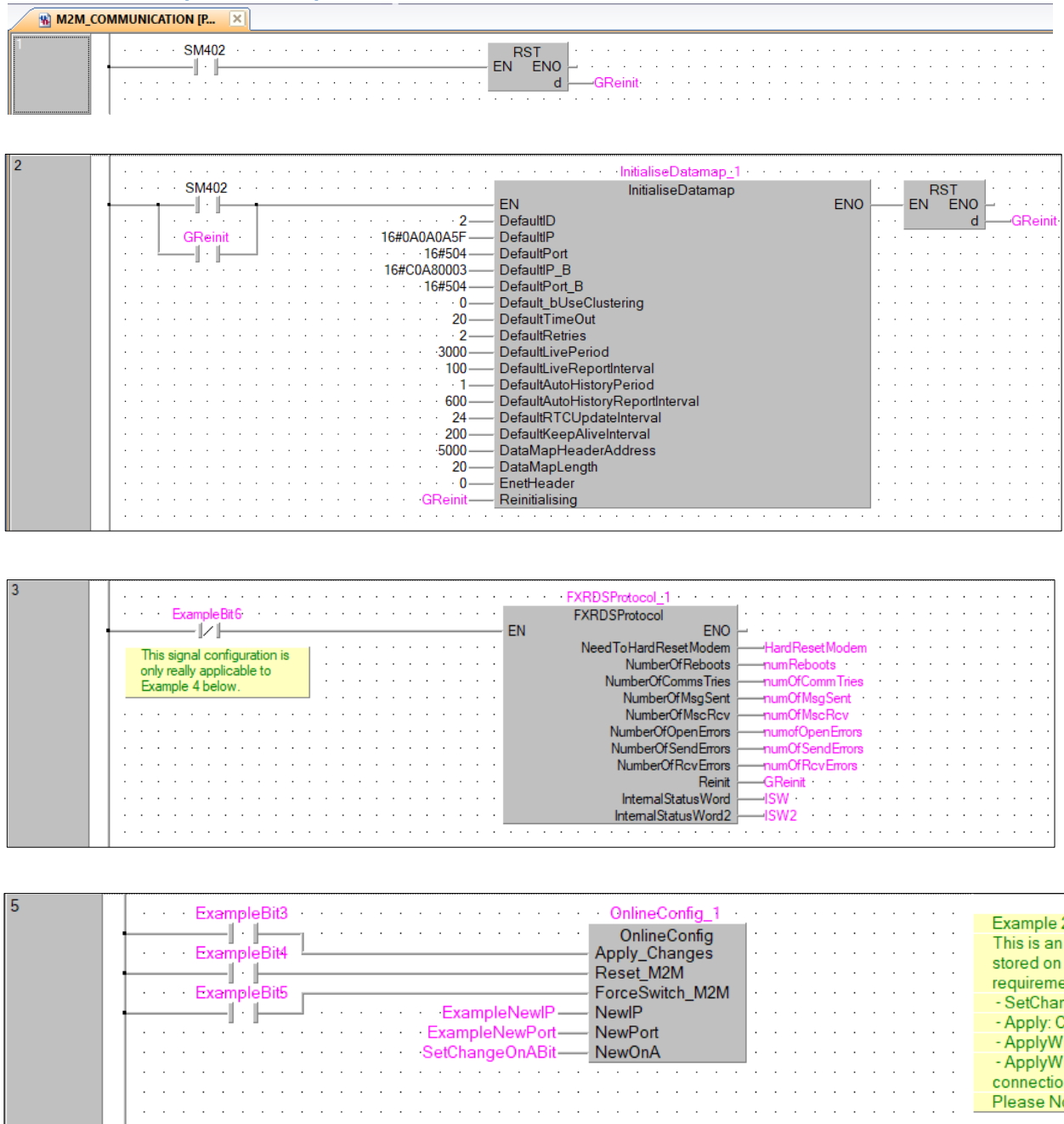
```
DataMap[2,0] := 16#0030;
DataMap[2,1] := 1350;
DataMap[2,2] := 16#FF00;
DataMap[2,3] := 16#00FF;
```

This is a 16-bit WORD variable that is being treated as 16 packed bits. These bits are being reported both historically and as unsolicited data. If any of the 8 most significant bits change, then the entire value will be reported as an unsolicited message. If any of the 8 least significant bits change then the entire value will be reported to the historical buffer. In the event of a race condition, the entire value will be reported both as an unsolicited update and as a historical message.

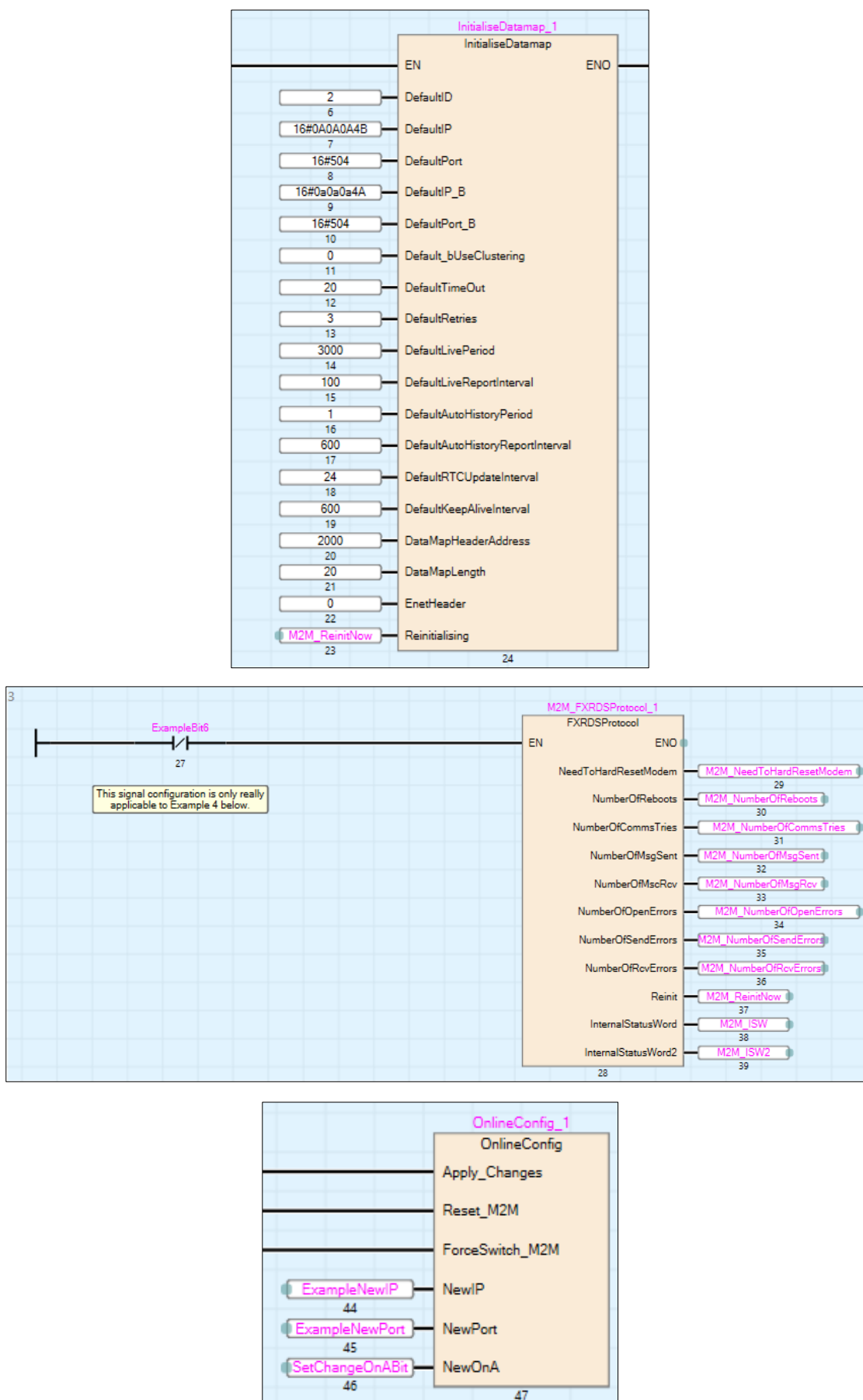
5.5.5. Communication Program

This program should be written in function block diagram. This is the program that initialises the communication and controls the functionality of the communications protocol.

GX Works 2: (QUDE PLC)



GX Works 3 (for FX5 PLC):

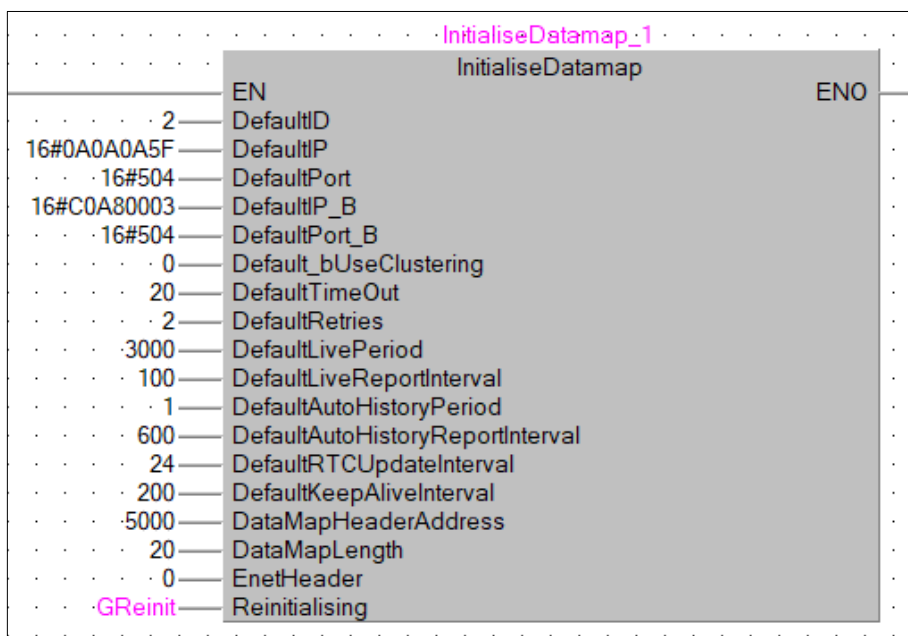


This program consists of basically three blocks.

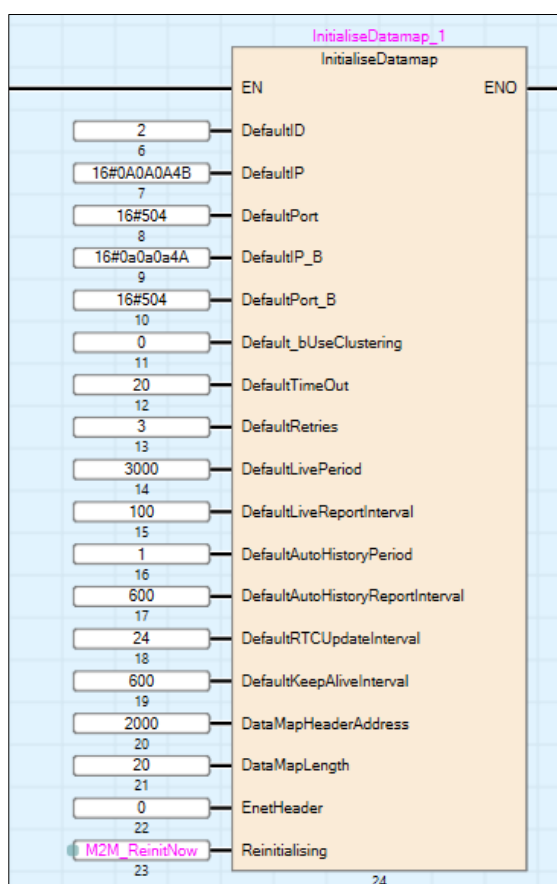
Initialise DataMap

The first of these blocks is the InitialiseDataMap block. This block is used to configure the initial connection parameters of the system and also to set various parameters that the library will use to ensure communication. This block should only be run once on PLC startup to ensure all communication parameters are properly initialised. Then the protocol block will ensure that if needed it will be rerun via the "initialising" input variable.

GX Works 2:



GX Works 3:



DefaultID: This is the station identifier that this unit will be initialised to if this variable in the PLC memory has been cleared. It is recommended that an initial valid ID is specified and that the PLC memory is cleared before the program is loaded.

DefaultIP: (hex) This is the primary (A) default Fixed Public IP address for the Adroit Master Server that the PLC will try and connect to if the PLC has no valid IP address specified in memory. It is recommended that an initial valid IP address is specified and that the PLC memory is cleared before the program is loaded. This IP address is made out of a byte per IP address field. For example, an IP with the value 16#C0A8000B converted to the regular Ipv4 format is 192.168.0.11.

*Translate from standard IP format to hex: 192 = **C0**, 168 = **A8**, 0 = **00**, 0B = **11***

DefaultPort: (hex) This is the primary default IP port that the PLC will try and connect on if the PLC has no port stored in the memory. It is recommended that an initial valid port is specified and that the PLC memory is cleared before the program is loaded.

DefaultIP_B: (hex) This is the secondary (B) default IP address that the PLC will try and connect on if the system is configured to be on Dual-port mode and the connection fails to establish on the primary connection. This will be connected on if there is no valid IP address for the secondary connection stored in the PLC memory. It is recommended that an initial valid IP is specified and the PLC memory is cleared before the program is loaded.

DefaultPort_B: (hex) This is the secondary (B) default IP address that the PLC will try and connect on if the system is configured to be on Dual-port mode and the connection fails to establish on the primary connection. This and the DefaultIP_B will be connected on if there is no valid IP address and port for the secondary connection stored in the PLC memory. It is recommended that an initial valid port is specified and the PLC memory is cleared before the program is loaded.

Default_bUseClustering: This is the default setting for the Standalone or Dual-port mode configuration of the system. If this value is 0, the default for the system is configured to be in Standalone, while a value of 1 configures the system to be in Dual-port mode – allowing usage of the secondary connection details. This will be used in the system if a valid previously defined setting has not been stored in the PLC memory. It is recommended that an initial setting is defined and the PLC memory is cleared before the program is loaded.

DefaultTimeOut: This is the timeout that the PLC will wait for messages before assuming that the message did not reach its end point and retransmitting. This value is in hundreds of milliseconds, therefore the illustrated value of 50 can be translated as 5 seconds.

DefaultRetries: This is the number of times that the PLC will attempt to send certain messages which are not critical, before it abandons the attempts. Critical messages like unsolicited updates, will not follow this rule and will continue to be transmitted until acknowledged by the SCADA system.

DefaultLivePeriod: This is the time that the PLC will stay in live mode and send continues updates to the SCADA system. This timer is configured with 100mS resolution. Thus the value of 3000 illustrated here can be interpreted as 300 seconds or 5 minutes.

DefaultLiveReportInterval: This is the interval at which the PLC will send complete system snapshots to the SCADA system. This timer is configured with 100mS resolution. Thus the value of 100 illustrated here will mean that a complete snapshot of the system will be sent every 10 seconds.

DefaultAutoHistoryPeriod: This value determines how long the PLC will continue to send automatic history updates after receiving the Auto Report History command from the SCADA. This

value has a hourly resolution and thus the value illustrated here means that the PLC will continue to send automatic history updates for one hour after receiving the command.

DefaultAutoHistoryReportInterval: This value determines how often the PLC will send automatic history update messages after receiving the Auto Report History command from the SCADA. This value has a 100mS resolution so the illustrated value of 600 means that updates will be sent every 60 seconds or one minute.

DefaultRTCUpdateInterval: This value governs how often the PLC will ask the SCADA to update its time. This value has a hourly resolution. In this example the PLC will thus raise the NeedTime status flag once a day to prompt the SCADA to update the Real time clock of the PLC.

DefaultKeepAliveInterval: This value governs how often the PLC will send a keep alive message to the SCADA to ensure that it has not lost communication. These keep alive messages are important to not only ensure that the connection stays active, but to notify the SCADA of any changes in the status of the PLC communication. This timer has a 100mS resolution timer and thus the illustrated value of 200 results in keep alive messages being sent every 20 seconds.

DataMapHeaderAddress: The address of the device where the DataMap that was defined in the configuration program starts.

DataMapLength: The length in number of items that the DataMap comprises.

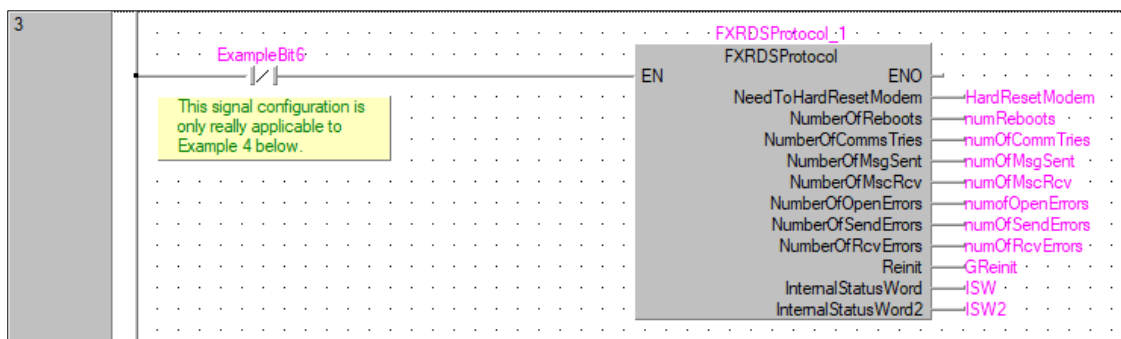
EnetHeader: The header address of the enet module. This is the same address as used by the FX Configurator EN program to configure the E-net Module.

Reinitialising: Added in rev613 to overcome certain lockup conditions. This is driven from the **Reinit** output var of the protocol block. It is used to restore most of the runtime parameters to the E-net module.

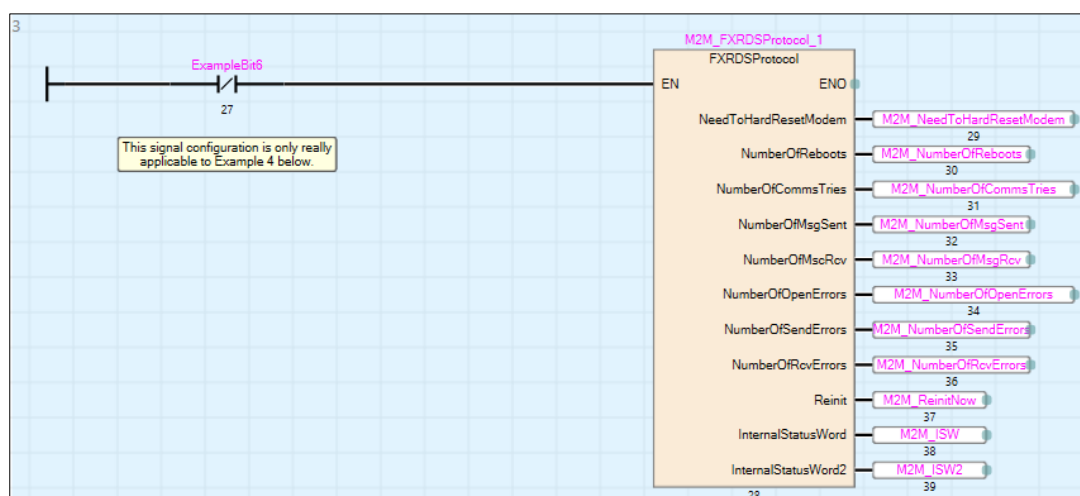
FXRDSProtocol

The FXRDSProtocol function block contains the core implementation of the protocol and is the part of the library that will ensure that communication takes place successfully. To assist in the monitoring of the communication the block exposes a number of variables that can be used for monitoring communication.

GX Works 2: (for QUDE PLC)



GX Works 3 (for FX5 PLC):



The FXRDSProtocol function block contains the core implementation of the protocol and is the part of the library that will ensure that communication takes place successfully. To assist in the monitoring of the communication the block exposes a number of variables that can be used for monitoring communication.

NeedToHardResetModem: This output will be switched on when the library has been out of communication for a while and switched off again after 5 seconds. This can be used to drive an output on the PLC to cycle the power to the modem used for communication. It is recommended that this output be configured and utilised to ensure communication recovers in the case that the modem gets hung up in a unusable state. The modem will be reset after 5 min of no communication, then again after 10min of no communication, and again after 15 min and so forth until a threshold of one hour is reached. Once this threshold is reached, the modem will continuously be reset at one hour intervals until communication is restored.

NumberOfReboots: This variable will keep track of the number of times the NeedToHardResetModem output has come on since the last time the statistics was reset from the SCADA.

NumberOfCommsTries: This variable will keep track of the number of times the library has told the E-net module to stop and restart trying to communicate to the configured IP Address.

NumberOfMsgSent: This variable will keep track of the number of messages successfully sent from the PLC to the SCADA.

NumberOfMscRcv: This variable will keep track of the number of messages successfully received from the SCADA.

NumberOfOpenErrors: This variable will keep track of the number open errors reported by the E-net module. This number will generally be higher than the NumberOfCommsTries variable as the the E-net module will typically try more than once in the same time we allow for communications to be established.

NumberOfSendErrors: This variable will keep track of the number of Sending errors reported by the E-net module. A rise in this value will indicate a bad ethernet connection to the SCADA system.

NumberOfRcvErrors: This variable will keep track of the number of receiving errors reported by the E-net module. A rise in this value will indicate a bad ethernet connection to the SCADA system.

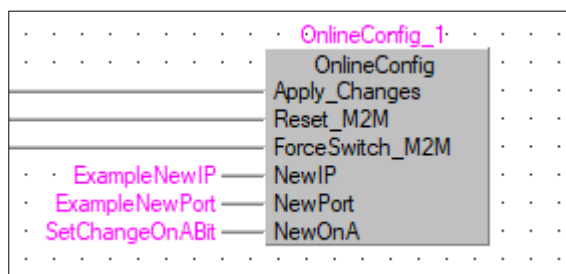
Reinit: This output variable will be used to rerun the above InitialiseDataMap FB above if required.

InternalStatusWORD: This is identical to SW0 available on the master, it can be used via bitstripping to determine what modes are active, for use outside the communications blocks. For example while in LIVE mode the PLC may fetch data faster, etc.

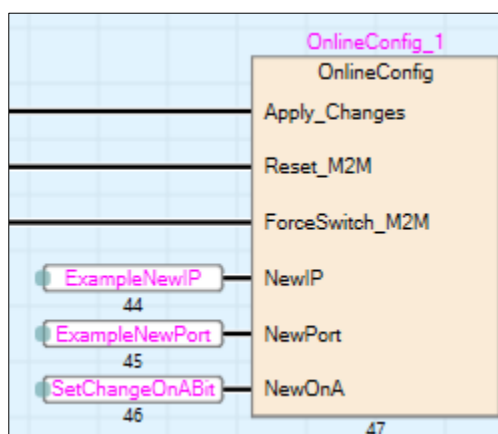
InternalStatusWORD2: *(Available with Q rev622, FX5U rev634 or later)* This is identical to SW2 available in the Config Area, Config WORD 29. This contains the flags regarding the dual-port mechanism, indicating which connection is currently being used and if a reset or force switch is required to bring in new changes. This will be defined in more detail later on in this document.

OnlineConfig

GX Works 2:



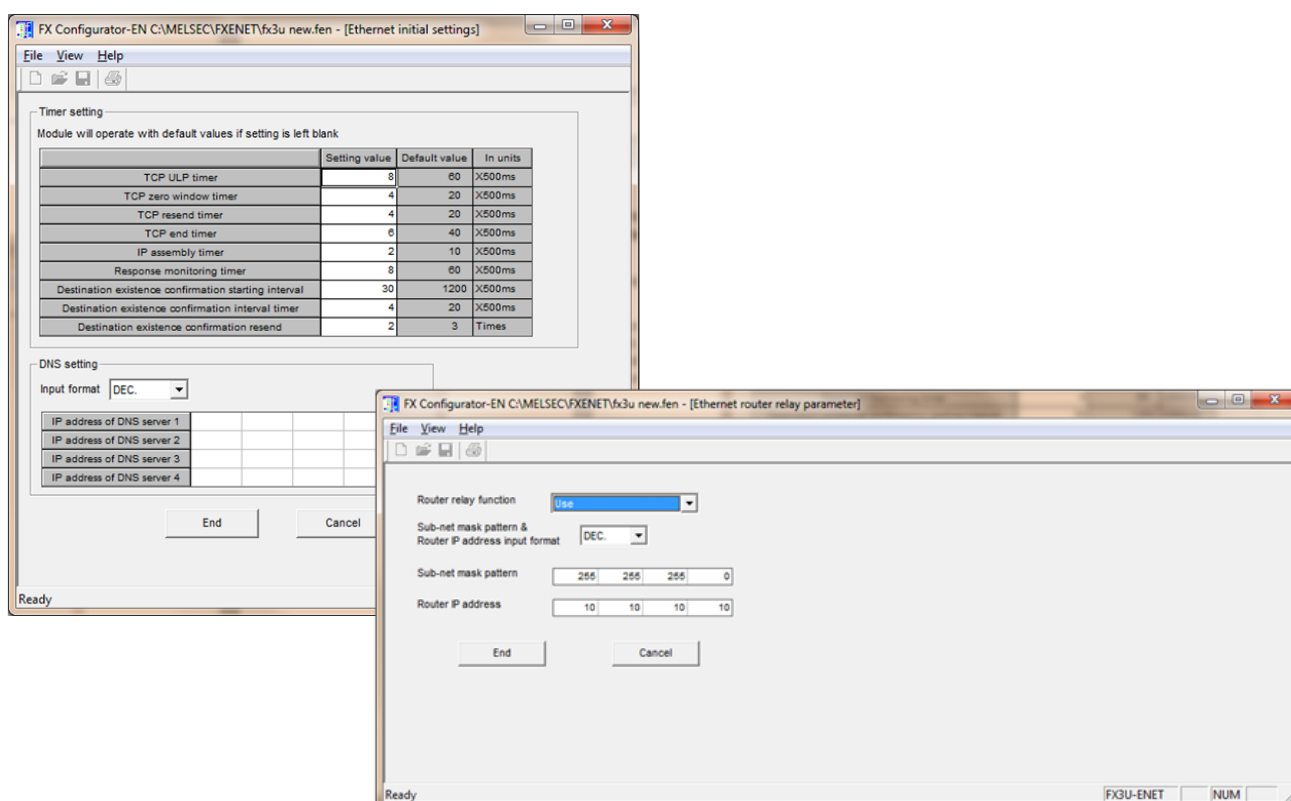
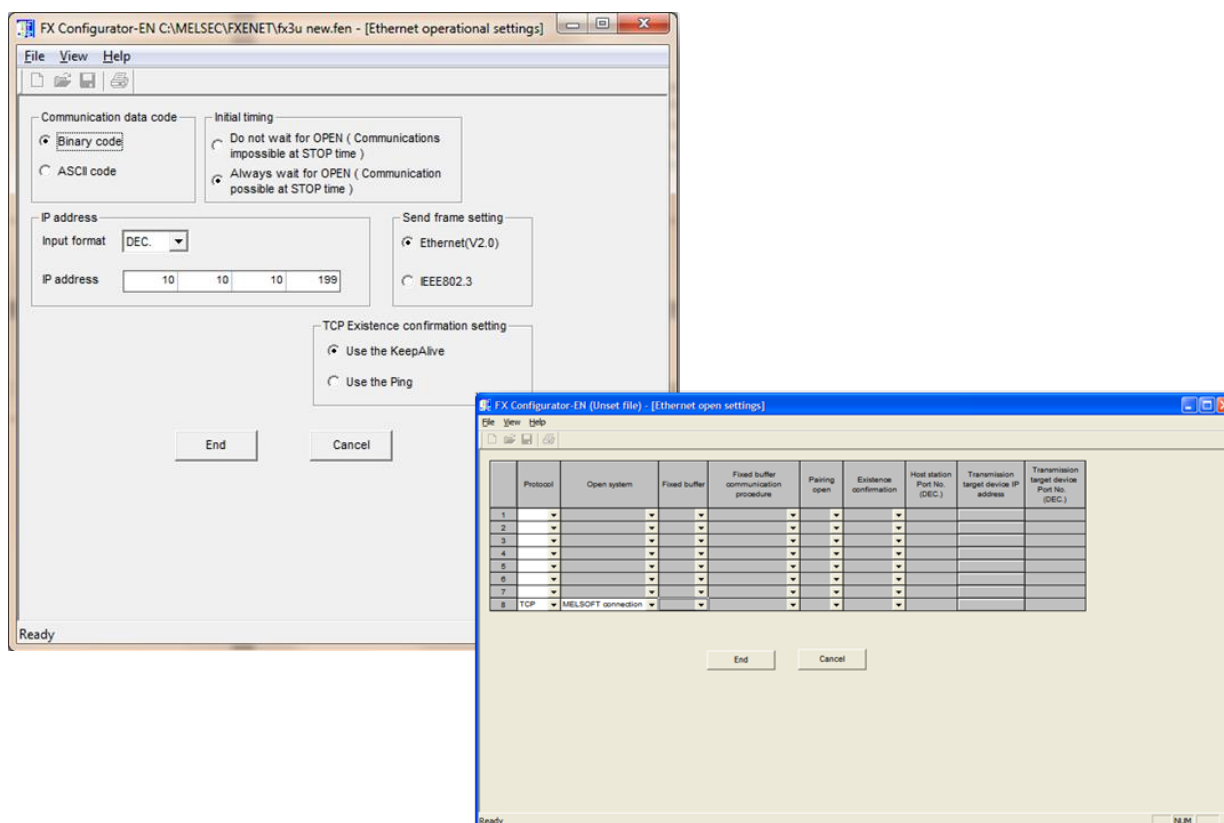
GX Works 3 (for FX5 PLC):



As of Q-Series rev620 and the FX5 rev634, this function block will allow the user to change the IP and Port details that the PLC will use to communicate on the fly for both primary and secondary connections, while also allowing the user to perform a Reset action and a ForceSwitch action (note: changes do not have to be made to be able to use the Reset_M2M and ForceSwitch_M2M features). Simply write the **NewIP** and **NewPort** values to the function block and pulse the **Apply_Changes** input. These values will then be loaded into buffers on the PLC. Depending the current system mode of operation (Standalone or Dual-port), the changes applied and the desired action to be taken, the user must then cause a Reset (generally for maintaining the same connection, primary or secondary, that we are currently on) by pulsing the **Reset_M2M** line, or a Force Switch (for switching the connection between primary and secondary and vice versa) by pulsing the **ForceSwitch_M2M**. To select what connection details are to be changed, the use of the **NewOnA** line is available. If this line is held high (1) then any changes when the **Apply_Changes** is pulsed will affect the primary (A). If this line is held low (0) then any changes when the **Apply_Changes** is pulsed will affect the secondary (B). Failure to connect based on the number of retries set in the system, the connection will restore last known good connections.

5.6. FX Configurator EN for FX3U

The following configuration is recommended for the E-net Module. **The first two connections on the E-net module must be reserved for this protocol.**



5.7. Communications Setup for QUDE and L-Series

Built-in Ethernet Port Open Setting

IP Address/Port No. Input Format: DEC

	Protocol	Open System	TCP Connection	Host Station Port No.	Destination IP Address	Destination Port No.
1	TCP	Socket Communication	Active	1284	10. 10. 10. 32	1284
2	TCP	MELSOFT Connection				
3	TCP	MELSOFT Connection				
4	TCP	MELSOFT Connection				
5	TCP	MELSOFT Connection				
6	TCP	MELSOFT Connection				
7	TCP	MELSOFT Connection				
8	TCP	MELSOFT Connection				
9	TCP	MELSOFT Connection				
10	TCP	MELSOFT Connection				
11	TCP	MELSOFT Connection				
12	TCP	MELSOFT Connection				
13	TCP	MELSOFT Connection				
14	TCP	MELSOFT Connection				
15	TCP	MELSOFT Connection				
16	TCP	MELSOFT Connection				

(*) IP Address and Port No. will be displayed by the selected format.
Please enter the value according to the selected number.

End Cancel

Q Parameter Setting

PLC Name | PLC System | PLC File | PLC RAS | Boot File | Program | SFC | Device | I/O Assignment | Multiple CPU Setting | **Built-in Ethernet Port Setting**

IP Address Setting

Input Format: DEC

IP Address: 10 10 10 210

Subnet Mask Pattern: 255 255 255 0

Default Router IP Address: 10 10 10 1

Open Setting

FTP Setting

Time Setting

Communication Data Code

☒ Binary Code

☐ ASCII Code

☐ Enable online change (FTP, MC Protocol)

☐ Disable direct connection to MELSOFT

☐ Do not respond to search for CPU (Built-in Ethernet port) on network

IP packet transfer setting

IP packet transfer setting

Set if it is needed(Default / Changed)

Print Window... Print Window Preview Acknowledge XY Assignment Default Check End Cancel

Q Parameter Setting

PLC Name | PLC System | PLC File | PLC RAS | Boot File | Program | SFC | **Device** | I/O Assignment | Multiple CPU Setting | Built-in Ethernet Port Setting

	Sym.	Dig.	Device Points	Latch (1) Start	Latch (1) End	Latch (2) Start	Latch (2) End	Local Device Start	Local Device End
Input Relay	X	16	8K						
Output Relay	Y	16	8K						
Internal Relay	M	10	8K						
Latch Relay	L	10	8K						
Link Relay	B	16	8K						
Annunciator	F	10	2K						
Link Special	SB	16	2K						
Edge Relay	V	10	2K						
Step Relay	S	10	8K						
Timer	T	10	2K						
Retentive Timer	ST	10	0K						
Counter	C	10	1K						
Data Register	D	10	12K						
Link Register	W	16	8K						
Link Special	SW	16	2K						
Index	Z	10	20						

Device Total: 28.8 K Words
Word Device: 25.0 K Words
Bit Device: 44.0 K Bits

The total number of device points is up to 29 K words.
Latch(1) : Able to clear the value by using a latch clear.
Latch(2) : Unable to clear the value by using a latch clear. Clearing will be executed by remote operation or program.
Scan time is extended by the latch range setting (including L).
If the latch is necessary, please set the required minimum latch range.
When using the local devices, please do the file setting at PLC file setting parameter.

File Register Extended Setting
Capacity: 16 K Points

	Sym.	Dig.	Device Points	Latch (1) Start	Latch (1) End	Latch (2) Start	Latch (2) End	Device No. Start	Device No. End
File Register	ZR(R)	10	16K					ZR0	ZR16383
Extended Data	D	10	0K						
Extended Link	W	16	0K						

Following setting are available when select "Use the following file" in file register setting of PLC file setting.
-Change of latch(2) of file register.
-Assignment to expanded data register/expanded link register of a part of file register area.

Indexing Setting for ZR Device
32Bit Indexing
☒ Use Z Z After (0 -- 18)
☐ Use ZZ

Print Window... | Print Window Preview | Acknowledge XY Assignment | Default | Check | End | Cancel

Q Parameter Setting

PLC Name | PLC System | **PLC File** | PLC RAS | Boot File | Program | SFC | Device | I/O Assignment | Multiple CPU Setting | Built-in Ethernet Port Setting

File Register
☐ Not Used
☐ Use the same file name as the program
Corresponding Memory: []
☒ Use the following file
Corresponding Memory: Standard RAM (Drive 3)
File Name: MAIN
Capacity: 16 K Points (1K--4086K Points)
☐ Transfer to Standard ROM at Latch data backup operation.
Following settings are available in device setting when select "Use the following file" and specify capacity.
-Change of latch(2) of file register.
-Assignment to expanded data register/expanded link register of part of file register area.

Device Initial Value
☒ Not Used
☐ Use the same file name as the program
Corresponding Memory: []
☐ Use the following file
Corresponding Memory: []
File Name: []

File for Local Device
☒ Not Used
☐ Use the following file
Corresponding Memory: []
File Name: []

File used for SP.DEVST/S.DEVLD Instruction
☒ Not Used
☐ Use the following file
Corresponding Memory: []
File Name: []
Capacity: [] K Points (1K--512K Points)

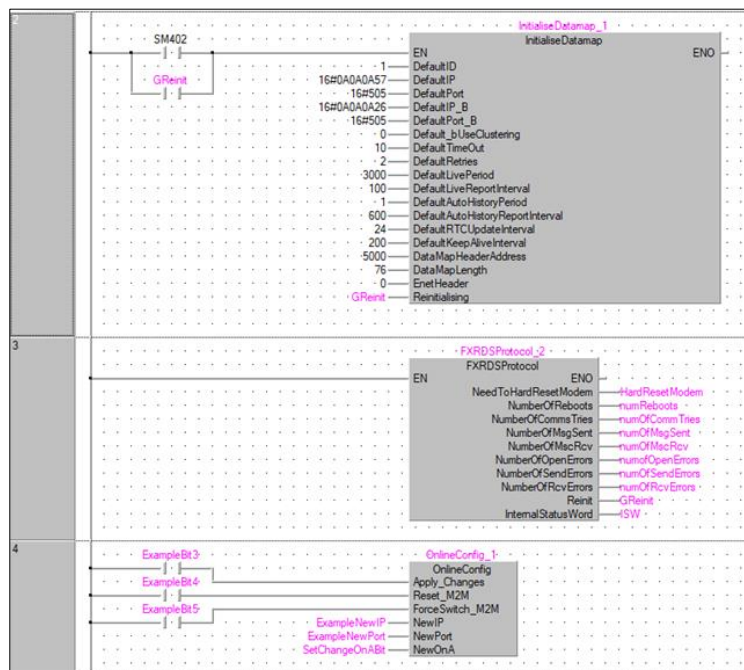
Comment File Used in a Command
☒ Not Used
☐ Use the same file name as the program
Corresponding Memory: []
☐ Use the following file
Corresponding Memory: []
File Name: []

Print Window... | Print Window Preview | Acknowledge XY Assignment | Default | Check | End | Cancel

The PLC program is named QUDE_M2M.gxw

In the project there is a **Communication** program in the Scan process and **BuildDataMap** in the Event process (once off at startup).

SUGGESTION: Take this GX Works2 project, make a copy and rename it to what you want to call your project. Create your program using this as a template. Now you do not need to copy and paste and make sure the labels and FB's are correct, as it is already in the program.



```

BuildDataMap [PRG] Program...
DataMap[0,0] := 16#0011;
DataMap[0,1] := 1000;
DataMap[0,2] := 16#0002;
DataMap[0,3] := 16#0005;

DataMap[1,0] := 16#0011;
DataMap[1,1] := 1001;
DataMap[1,2] := 16#0002;
DataMap[1,3] := 16#0005;

DataMap[2,0] := 16#0011;
DataMap[2,1] := 1002;
DataMap[2,2] := 16#0005;
DataMap[2,3] := 16#0005;

DataMap[3,0] := 16#0001;
DataMap[3,1] := 1003;
DataMap[3,2] := 16#001A;
DataMap[3,3] := 16#0005;

```

Class	Label Name	Data Type	Constant	Device	Address
1	VAR_GLOBAL	DataMap	Word(Signed)(0..9,0..3)	D1400	%MW0.1400
2					
3					

DefaultID: This is the station identifier that this unit will be initialized to if this variable in the PLC memory has been cleared. This ID must be unique to each M2M device as this is the differentiator for Adroit.

DefaultIP: This is the primary (A) IP Address that the PLC will try and connect to. This is the Fixed Public IP address for the Adroit Master server. **16#0A0A0A57 Hex = 10.10.10.87 IPv4 = 168430167 Dec.** The Fixed Public IP address for this example is **192.168.100.9**.

DefaultPort: This is the primary IP Port that the PLC will try and connect to the Adroit server. **16#505 Hex = 1285 Dec.** So, the port in Adroit is **1285**.

DefaultIP B: This is the secondary (B) IP Address that the PLC will try and connect to when in Dual-port mode and the primary connection has failed. This is the Fixed Public IP address for the Adroit Standby server. **16#0A0A0A26 Hex = 10.10.10.38 IPv4 = 168430118 Dec.** The Fixed Public IP address for this example is **192.168.100.10**.

DefaultPort B: This is the secondary IP Port that the PLC will try and connect to the Adroit server when on Dual-port mode and the primary connection has failed. **16#505 Hex = 1285 Dec.** So, the port in Adroit is **1285**.

Default bUseClustering: This is set when the system has a Dual-port configuration, reset when the system has Standalone configuration. Both Masters need to be contactable with fixed IP addresses.

DefaultTimeout: This is the timeout that the PLC will wait for messages before assuming that the message did not reach its end point and retransmitting.

DefaultRetries: This is the number of times that the PLC will attempt to send certain messages which are not critical, before it abandons the attempts. Critical messages like unsolicited updates, will not follow this rule and will continue to be transmitted until acknowledged by the SCADA system.

DefaultLivePeriod: This is the time that the PLC will stay in live mode and send continues updates to the SCADA system.

DefaultLiveReportInterval: This is the interval at which the PLC will send complete system snapshots to the SCADA system.

DefaultAutoHistoryPeriod: This value determines how long the PLC will continue to send automatic history updates after receiving the Auto Report History command from the SCADA.

DefaultAutoHistoryReportInterval: This value determines how often the PLC will send automatic history update messages after receiving the Auto Report History command from the SCADA.

DefaultRTCUpdateInterval: This value governs how often the PLC will ask the SCADA to update its time.

DefaultKeepAliveInterval: This value governs how often the PLC will send a keep alive message to the SCADA to ensure that it has not lost communication. These keep alive messages are important to not only ensure that the connection stays active, but to notify the SCADA of any changes in the status of the PLC communication.

DataMapHeaderAddress: The address of the device where the DataMap that was defined in the configuration program starts. This needs to be the same as specified in the *Global_M2M_DataMap*.

DataMapLength: The length in number of items/registers in the DataMap that is to be scanned to the Adroit Agent Server. By program default, its 20, but the maximum addresses allowed is 200 for 16-bit WORDs and 100 for 32-bit WORDs.

EnetHeader: Default 0 for internal Ethernet. The header address of the ENET module.

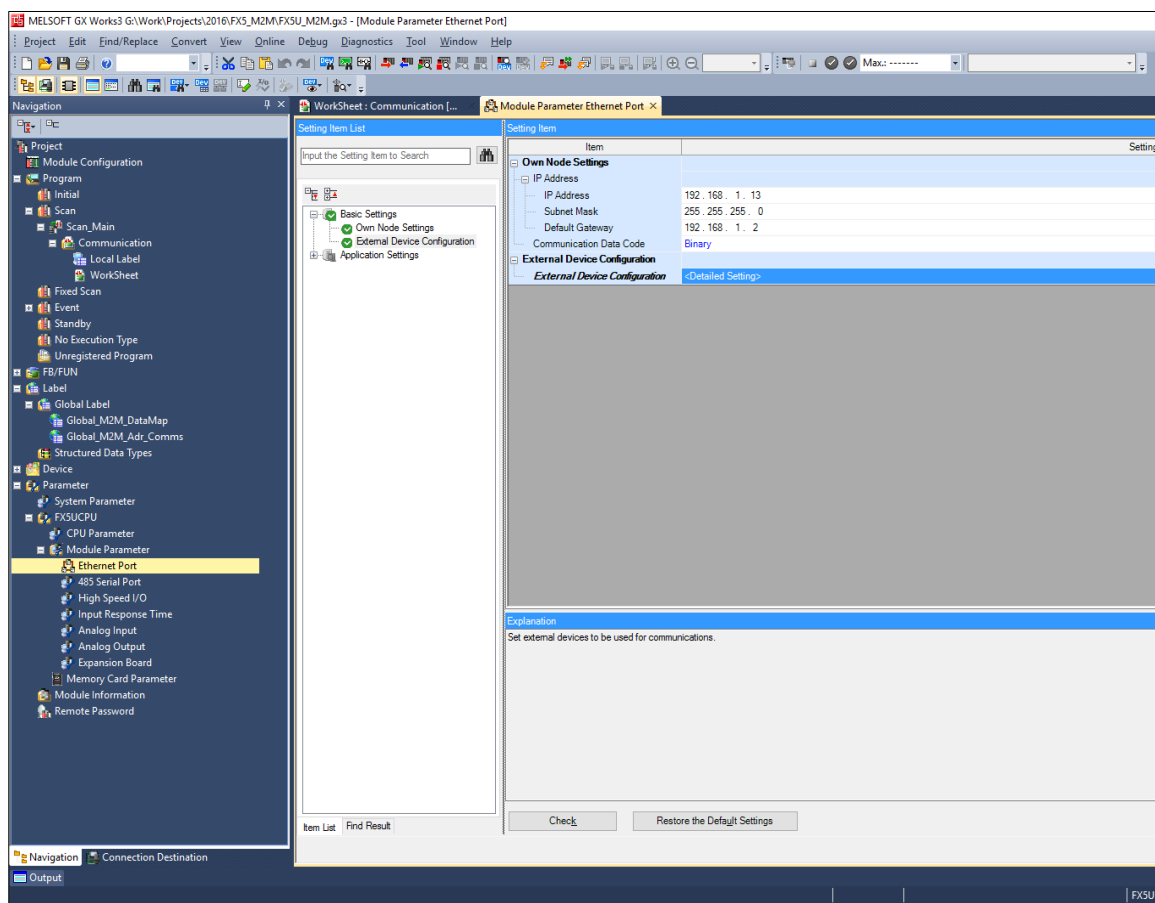
In the other 2 function blocks, no changes are needed.

BuildDataMap Program:

This is where the format of the PLC data going to SCADA is handled.

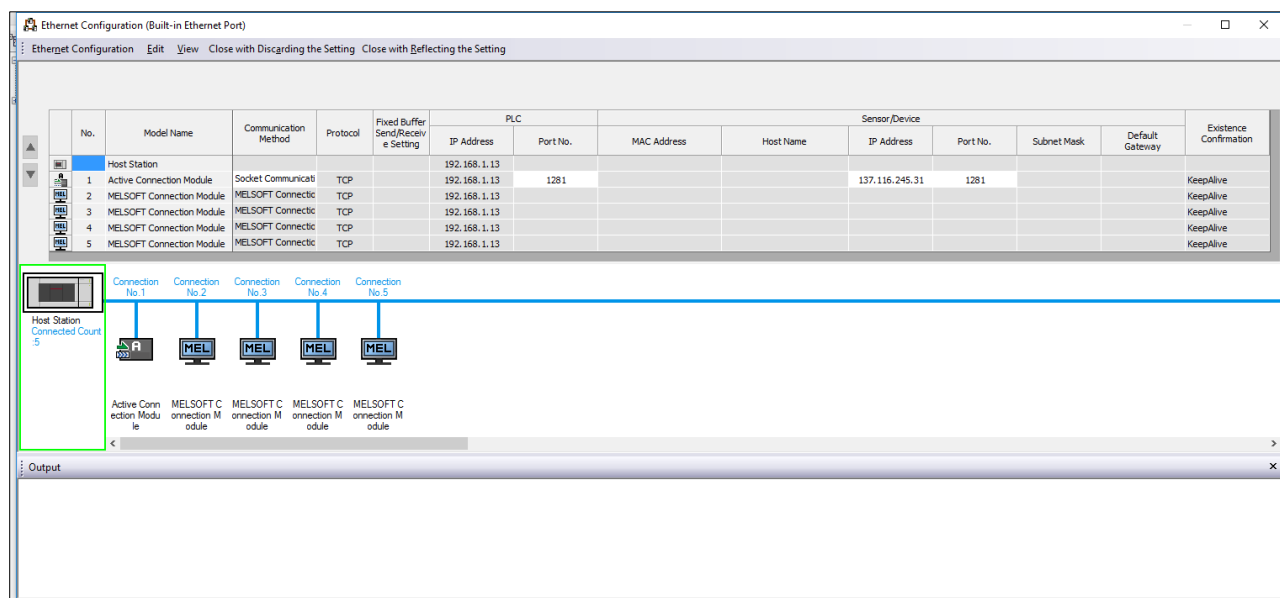
Example: D1000 a 16-bit value or 16-bit WORD containing packed signals. In Adroit, the latter will be used for Marshal Agents. Also, whether the data is to be treated as unsolicited or as historical. The setup for this will be done in this DataMap. These addresses do not need to be D1000 but can be your existing program addresses. You just need to specify them here. The maximum scan points to scan to the SCADA is 200.

5.8. Communications Setup for FX5-Series



PLEASE NOTE

Default Gateway is the Modem IP address.





PLEASE NOTE

Sensor/Device IP address is the Fixed Public IP address to where to connect. Port number is the Port specified in Adroit Device Setup.

MELSOFT GX Works3 G:\Work\Projects\2016\FX5_M2M\FX5U_M2M.gx3 - [FX5UCPU CPU Parameter]

Project Edit Find/Replace Convert View Online Debug Diagnostics Tool Window Help

Navigation Worksheet: Communication [...] Module Configuration FX5UCPU CPU Parameter

Setting Item List

Input the Setting Item to Search

Setting Item

Item	Symbol	Device		Latch (I)	
		Points	Range		
Input	X	1024	0 to 1777		
Output	Y	1024	0 to 1777		
Internal Relay	M	7680	0 to 7679	Setting	No Setting
Link Relay	B	256	0 to FF	No Setting	No Setting
Special Link Rela SB		256	0 to FF		
Annunciator	F	128	0 to 127	No Setting	No Setting
Step Relay	S	4096	0 to 4095	Setting	
Timer	T	512	0 to 511	No Setting	No Setting
Retentive Timer	ST	16	0 to 15	Setting	No Setting
Counter	C	256	0 to 255	Setting	No Setting
Long Counter	LC	64	0 to 63	Setting	No Setting
Data Register	D	8000	0 to 7999	Setting	No Setting
Latch Relay	L	7680	0 to 7679		
Area Capacity			12.0K Word		
Total Device			11.1K Word		
Total Word Device			10.2K Word		
Total Bit Device			15.7K Bit		

Explanation

Input (X). Device points is 1024 points (fixed).

[Type] Bit Device, [Notation] OCT

Check Restore the Default Settings

MELSOFT GX Works3 G:\Work\Projects\2016\FX5_M2M\FX5U_M2M.gx3 - [FX5UCPU CPU Parameter]

Project Edit Find/Replace Convert View Online Debug Diagnostics Tool Window Help

Navigation: Project, Module Configuration, Program, Initial, Scan, Scan_Main, Communication, Local Label, Worksheet, Fixed Scan, Event, Standby, No Execution Type, Unregistered Program, FB/FUN, Label, Global Label, Global_M2M_DataMap, Global_M2M_Adr_Comms, Structured Data Types, Device, Parameter, System Parameter, FX5UCPU, CPU Parameter, Module Parameter, Ethernet Port, 485 Serial Port, High Speed I/O, Input Response Time, Analog Input, Analog Output, Expansion Board, Memory Card Parameter, Module Information, Remote Password

Setting Item List: Input the Setting Item to Search

- Name Setting
 - Operation Related Setting
 - RUN Contact Setting
 - Remote Reset Setting
 - Clock Related Setting
 - Interrupt Setting
 - Fixed Interval Setting
 - Fixed Scan Execution Mode Setting
 - Interrupt Priority Setting from Module
 - Service Processing Setting
 - Device/Label Access Service Processing Setting
 - File Setting
 - Initial Value Setting
 - Memory/Device Setting
 - Device/Label Memory Area Setting
 - Index Register Setting
 - Pointer Setting
 - RAS Setting
 - Scan Time Monitoring Time (WDT) Setting
 - Constant Scan Setting
 - Error Detections Setting
 - CPU Module Operation Setting at Error Detected
 - LED Display Setting
 - Program Setting
 - Program Setting
 - FB/FUN File Setting

Setting Item

Item	Symbol	Points	Range	Latch (T)	
File Register	R	8000	0 to 7999	No Setting	No Setting
Link Register	W	0		No Setting	No Setting
Link Special Register SW		0			
Area Capacity			8.0K Word		
Total Device			7.9K Word		
Total Word Device			7.9K Word		
Total Bit Device			0.0K Bit		

Explanation

Set device points and latch range used in file register (R).
(Note to set latch when option battery setting is mounted.)

[Type] Word Device, [Notation] DEC
(Setting range)
Device Points: 0 to 32768 [Points](4 points unit)

Check Restore the Default Settings

MELSOFT GX Works3 G:\Work\Projects\2016\FX5_M2M\FX5U_M2M.gx3 - [FX5UCPU CPU Parameter]

Project Edit Find/Replace Convert View Online Debug Diagnostics Tool Window Help

Navigation: Project, Module Configuration, Program, Initial, Scan, Scan_Main, Communication, Local Label, Worksheet, Fixed Scan, Event, Standby, No Execution Type, Unregistered Program, FB/FUN, Label, Global Label, Global_M2M_DataMap, Global_M2M_Adr_Comms, Structured Data Types, Device, Parameter, System Parameter, FX5UCPU, CPU Parameter, Module Parameter, Ethernet Port, 485 Serial Port, High Speed I/O, Input Response Time, Analog Input, Analog Output, Expansion Board, Memory Card Parameter, Module Information, Remote Password

Setting Item List: Input the Setting Item to Search

- Name Setting
 - Title Setting
 - Comment Setting
- Operation Related Setting
 - RUN Contact Setting
 - Remote Reset Setting
 - Clock Related Setting
- Interrupt Setting
 - Fixed Interval Setting
 - Fixed Scan Execution Mode Setting
 - Interrupt Priority Setting from Module
- Service Processing Setting
 - Device/Label Access Service Processing Setting
- File Setting
 - Initial Value Setting
- Memory/Device Setting
 - Device/Label Memory Area Setting
 - Index Register Setting
 - Pointer Setting
- RAS Setting
 - Scan Time Monitoring Time (WDT) Setting
 - Constant Scan Setting
 - Error Detections Setting
 - CPU Module Operation Setting at Error Detected
 - LED Display Setting
- Program Setting
 - Program Setting
 - FB/FUN File Setting

Setting Item

Item	
Device/Label Memory Area Setting	
Option Battery Setting	Not Mounted
Device/Label Memory Area Capacity Setting	
Device Area	
Device (high speed) Area Capacity	12 K Word
Device (standard) Area Capacity	8 K Word
Label Area	
Label/Latch Label Use Device Area Setting	Standard Area
Label Area Capacity	39 K Word
Latch Label Area Capacity	1 K Word
Device/Label Memory Configuration Confirmation	<Confirmation>
Check the Latch Configuration	<Confirmation>
Device/Label Memory Area Detailed Setting	
Device (high speed) Setting	<Detailed Setting>
Device (Standard) Setting	<Detailed Setting>
Latch type setting of the latch relay (L)	Latch (T)
Latch Label Latch Type	Latch (T)
Latch area of the latch label	Standard Latch Area
To use or not to use the routine timer of timer (T)	Not Use
Start device No. of routine timer of timer (T)	0
Index Register Setting	
Points Setting	
Total Points	24 Points
Index Register (Z)	20 Points
Long Index Register (LZ)	2 Points
Pointer Setting	
Total Points	4096 Points
Global Pointer	2048 Points
Pointer Type Label	2048 Points

Explanation

Please set to change the each area capacity of device/label memory.

Check Restore the Default Settings

6. General Notes and Troubleshooting

6.1. Contact Details

Email: support@adroit.co.za and drivers@adroit.co.za

Phone: +27 11 658-8100

Web: www.adroit.co.za

6.2. Adroit Technologies Knowledge Base

For additional information, please refer to our Knowledgebase website:

[Latest Drivers topics - Features, discussions, tips, tricks, questions, problems and feedback \(adroittechnologiesautomation.com\)](http://adroittechnologiesautomation.com)

6.3. Troubleshooting

1) Data area values change erratically back and forth between unsolicited / history updates and interrogate / manual / automatic polls (which fetch all data at once).

- Unsolicited and events may differ so make sure that this is perhaps not valid changes.
- If, however, an invalid pattern is identified that seems incorrect it may indicate a map mismatch between the PLC definition and the driver/scanning address definitions (see physical vs logical addressing discussion for DATA area above) because unsolicited updates reference logical indexes per event, where polls fetch the whole buffer which is unpacked by referencing physical position.
- Check the datascope, if a red message indicating a different session being switched to and or back and forth, it may be an indication of 2 PLC's with the same station ID on the same port.
(NB Station ID's must be unique on a port)

2) Changes to the secondary connection (b) are not being implemented and CW29 is not changing.

Change your mode of operation for Standalone to Dual-port mode. Any changes on B will then be reflected and the necessary bits in CW29 will reflect as such. Should this problem continue through this, please contact Adroit Support.

3) PLC device not connecting to the other Agent Server in a Cluster server configuration.

- If you are using Shadow Scanning, please note that no connection can be made by a PLC to a server that is configured as the Standby server. Please check that the server you are trying to connect to is the Master server. Failure to connect to the Master means there is a fault in the connection or that the server is unreachable to the PLC.
- If you are using Parallel Scanning, please note that your PLC can connect to any of the Agent Servers and that there is a different problem with your connection. Please check your connection between the device and the Agent Servers.

4) On the FX5 PLC, the error light indicator is on (ERR) but the program is still operating. What is causing this?

There are two known reasons for why this may occur with the MIT_M2M program. They are the following:

- [H1090] Battery error – either the battery unit for the PLC is completely missing or the battery is low. This may require getting a new battery unit.



PLEASE NOTE

Problems with the battery can lead to the latched memory in the PLC to lose its stored values when AC power is lost to the PLC, causing you to lose configuration data. This means you will lose details such as Last Known Good IP connection details and your dual-port configuration if it has changed from the default.

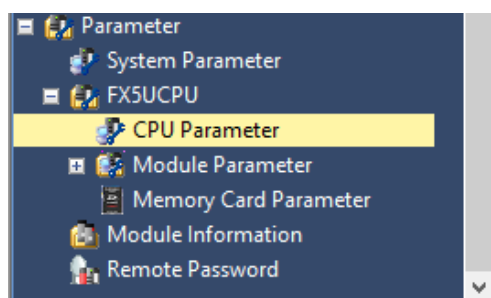
- [HCODE] Socket communication message send failed. This is a common error that occurs on a failed connection or connection timeout from the PLC. This is more indicative of connectivity issues experienced at some point in the program execution. Check your connection – if you are still connected, then clear the error.
- To check the error codes on the PLC, go to the menu bar and click on **Diagnostics**, then **Module Diagnostics (CPU diagnostics)** to find the error code.

5) When I try to build a program using the PLC library, I keep getting an error 0x1212101F – “Local latch label area capacity of program is insufficient. FB instance ensure reserved area in latch label area...”. How do I fix this issue?

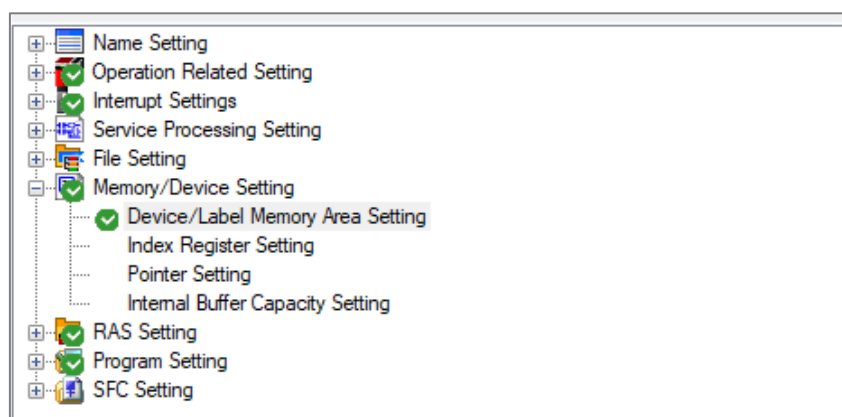
The issue here is the sizing of the local latch label area. The default is set to 1K as specified in the FX5U. For the FX5U, this can be increased to a maximum of 2K, which is what is required for MIT_M2M. Under the **MELSEC iQ-F FX5 User's Manual (Application) - Device/Label Memory Area Setting**, the memory areas are given as follows:

Item			Description	Setting range	Default
Option Battery Setting			Set when using option battery. The points which can be held can be increased by this setup. The latch device of standard area can be held with a battery. The latch area of latch label can be changed to battery latch area from standard latch area (nonvolatile memory).	• Not Mounted • Mounted	Not Mounted
Device/Label Memory Area Capacity Setting	Device Area	Device (high speed) Area Capacity	Set the capacity of device (high speed) area.	☞ Page 111 The Setting Range of the Capacity of Each Area	12 K word
		Device (standard) Area Capacity	Set the capacity of device (standard) area.	☞ Page 111 The Setting Range of the Capacity of Each Area	35 K word
	Label Area	Label/Latch Label Use Device Area Setting	Select the used device area of label and latch label from standard area and high speed area. When device (high speed) area + label area + latch label area is 12 K word or less, it is possible to set label area/latch label area in high-speed area.	• Standard Area • HighSpeed Area	Standard Area
		Label Area Capacity	Sets the capacity of the label area to be used for non-latched labels.	☞ Page 111 The Setting Range of the Capacity of Each Area	12 K word
		Latch Label Area Capacity	Sets the capacity of the latch label area to be used for latch-type labels.	☞ Page 111 The Setting Range of the Capacity of Each Area	1 K word
		Device/Label Memory Configuration Confirmation	Display the configuration of each area which set in device/label memory area capacity.	—	—
		Check the Latch Configuration			

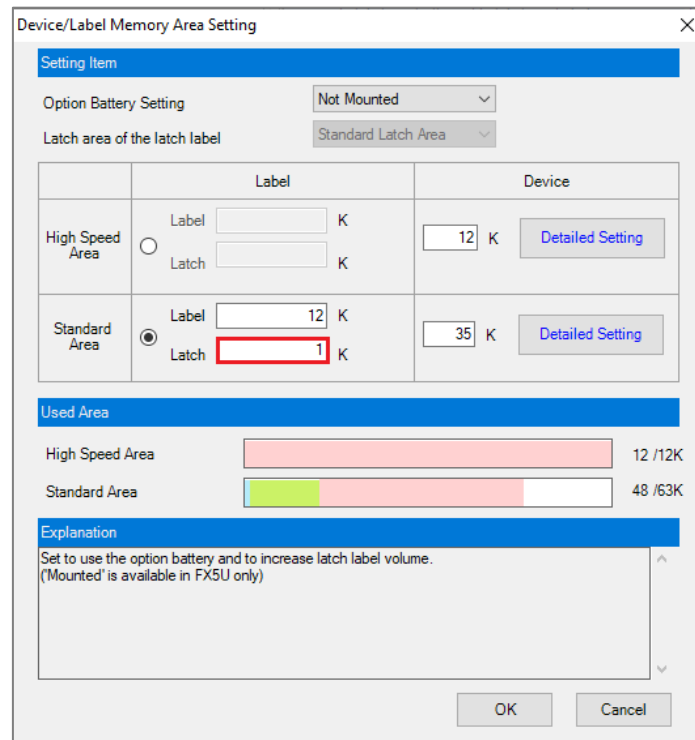
You will need to adjust the sizes of the areas in your program to accommodate this requirement. To do this, in GX Works3 got to the **Navigation** tree window and open **Parameter** -> **FX5UCPU** -> **CPU Parameter**



In the window that opens, navigate to **Memory/Device Setting** -> **Device/Label Memory Area Setting**.

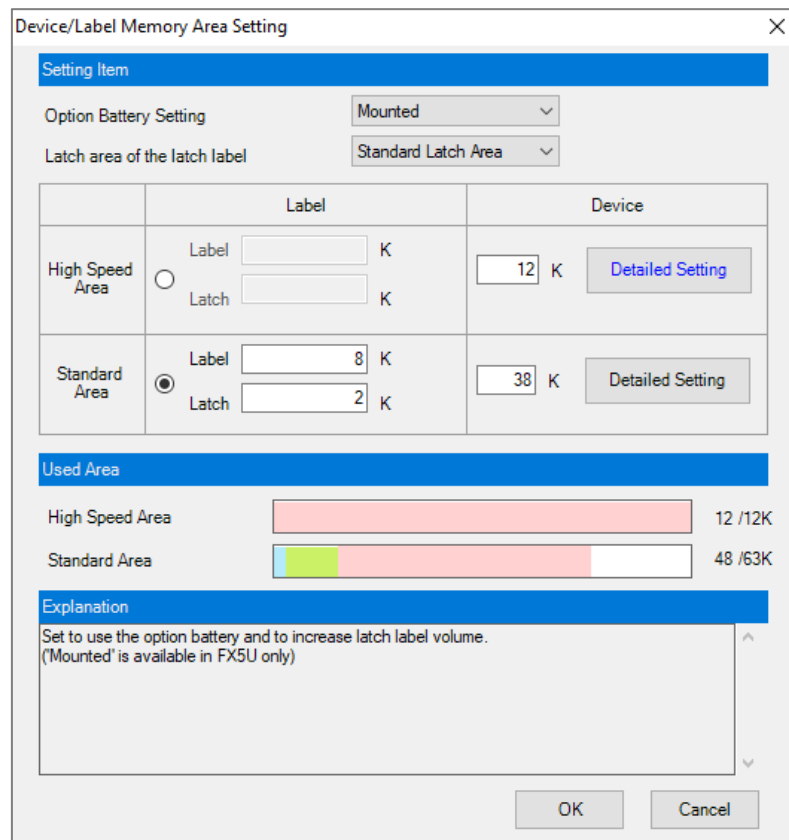


In the dialog that appears, increase the size of the Local Latch Area in from **1K** to **2K**.



The dialog box 'Device/Label Memory Area Setting' has a 'Setting Item' section with 'Option Battery Setting' set to 'Not Mounted' and 'Latch area of the latch label' set to 'Standard Latch Area'. Below is a table with two rows: 'High Speed Area' and 'Standard Area'. The 'Standard Area' row is selected with a radio button. In this row, the 'Label' is set to 12 K and the 'Latch' is set to 1 K (highlighted with a red box). The 'Device' column shows 35 K and a 'Detailed Setting' button. Below the table is a 'Used Area' section with two bars: 'High Speed Area' (12 / 12K) and 'Standard Area' (48 / 63K). An 'Explanation' section at the bottom states: 'Set to use the option battery and to increase latch label volume. (*Mounted* is available in FX5U only)'. 'OK' and 'Cancel' buttons are at the bottom right.

The following settings were working with the FX5U program:



The dialog box 'Device/Label Memory Area Setting' has 'Option Battery Setting' set to 'Mounted' and 'Latch area of the latch label' set to 'Standard Latch Area'. The 'Standard Area' row is selected. In this row, the 'Label' is set to 8 K and the 'Latch' is set to 2 K. The 'Device' column shows 38 K and a 'Detailed Setting' button. The 'Used Area' section shows 'High Speed Area' (12 / 12K) and 'Standard Area' (48 / 63K). The 'Explanation' section is the same as in the previous dialog. 'OK' and 'Cancel' buttons are at the bottom right.

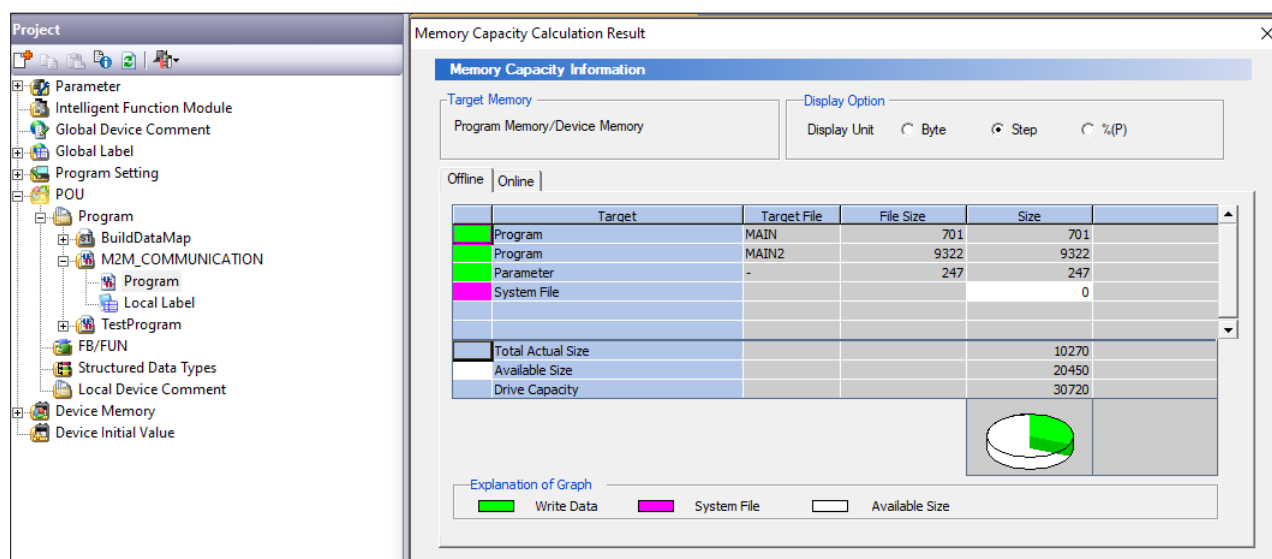
Bear in mind this for the Standard Area:

Label Area Capacity + Latch Label Area Capacity + Device (standard) Area Capacity ≤ 48 K Word (1 K word unit)

7. Appendix A – Memory Requirements

This section further elaborates on the memory requirements for the PLC utilizing the MIT_M2M protocol library and associated features.

GX Works 2:



For the Q/L-series driver, as of rev622, the program for the protocol itself as seen in MAIN2 in the image utilises 10270 Steps/41080 Bytes with all the associated example program features. The program for building the DataMap region of the protocol, in MAIN, will increase in size depending on the user requirements. For each element added to the DataMap and each increment of the DataMap size, 8 steps will utilised for MAIN (32 bytes of memory).

Keep this in mind for memory restrictions of the PLC being used.

FX5 Additional Memory Considerations:

- Number of Steps: 40 662 Steps required
- Labels used: 4880 (Local) + 68 (Global)
- Latch Label: 1344 (Local) [***Please read the following note***]



PLEASE NOTE

You may may need to increase your default size for this field from 1K to 2K.

8. Appendix B - Standalone/Dual-Port Details

Part of this revision is the updating of the dual-port system. As part of this revision, the user can now configure both the primary and secondary connection details and the way the system interacts with the driver using Adroit.

The idea is that the Dual-port mode of operation works as an alternative connection to connect on, should the primary connection fail. The user can also control when to switch over through a manual setting of AW0, bit 14 (dig14) called ForceSwitch. This will stop the current connection and bring in the alternative connection. Say for example the connection is on primary (A), a ForceSwitch will cause the connection to change from A to the secondary (B).

The following scenarios are key to understanding the system, for both Standalone and Dual-port modes of operation. These excerpts can be found in the 'M2M_Standalone and Dual-port Logic.pdf'.

8.1. On First Ever Boot-Up: Standalone

Step 1: System Overview

Hard written in PLC:
defaultA = [10.10.10.1; 1284]
defaultB = [10.10.10.2; 1285]

dest_conn = [0;0]

Primary (A)

A = [0;0]
LKG = [0;0]

Secondary (B)

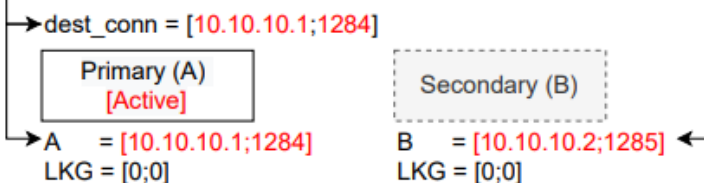
B = [0;0]
LKG_B = [0;0]

System in standalone does not rely on Secondary connection (B), though its connection details can still be altered.

The PLC needs to be provided with default IP addresses that are inserted into the program before it is compiled and written to the device. The system will identify that when booting up in this state, since Last Known Good (LKG) for both connection details are zero, we must load the default details.

Step 2: Loading First Connection

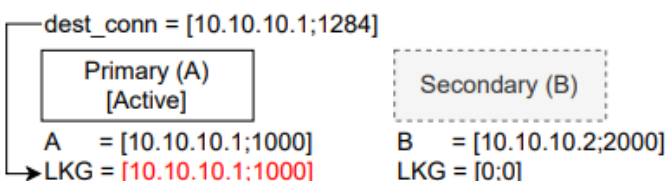
Hard written in PLC:
defaultA = [10.10.10.1; 1284]
defaultB = [10.10.10.2; 1285]



The system will load in the connection details for the primary connection (A) and the secondary connection (B)

Step 3: Connection Successful on Primary (A)

Hard written in PLC:
defaultA = [10.10.10.1; 1284]
defaultB = [10.10.10.2; 1285]



On the first connection on the Primary (A), the details of that connection will be stored in Last Known Good (LKG).

8.2. Change Connection Details [Maintaining the current connection]: Standalone

Step 1: System Overview

Hard written in PLC:

defaultA = [xxx; yyy]

defaultB = [zzz; uuu]

dest_conn = [10.10.10.1;1284]

Primary (A) [Active]	Secondary (B)
A = [10.10.10.1;1284]	B = [10.10.10.2;1285]
LKG = [10.10.10.1;1284]	LKG = [0;0]
NewA = [0; 0]	NewB = [0;0]

We are connected on A and we want to change the connection details of the Primary (A).

Step 2: New Connection Details Received

Hard written in PLC:

defaultA = [xxx; yyy]

defaultB = [zzz; uuu]

dest_conn = [10.10.10.1;1284]

Primary (A) [Active]	Secondary (B)
A = [10.10.10.1;1284]	B = [10.10.10.2;1285]
LKG = [10.10.10.1;1284]	LKG = [0;0]
NewA = [10.10.11.1;1300]	NewB = [0;0]

New connection details are received from the Agent Server (in this case since it is new A details, these are sent through on CW13, 14 and 15). Since we are in standalone, a Reset M2M flag will be raised (CW29, B1). From here, either a connection fail on A or a ResetM2M (AW0, B1) must be triggered by the operator for the change to be loaded into the system

Step 3: Connection details loaded

Hard written in PLC:

defaultA = [xxx; yyy]

defaultB = [zzz; uuu]

dest_conn = [10.10.11.1;1300]	
Primary (A) [Active]	Secondary (B)
A = [10.10.11.1;1300]	B = [10.10.10.2;1285]
LKG = [10.10.10.1;1284]	LKG = [0;0]
NewA = [10.10.11.1;1300]	NewB = [0;0]

On a connection fail or ResetM2M by the operator, the new details are loaded into A and the destination connection, where multiple connection attempts depending on the retries set by the system will take place.

Step 4a: Connection Successful

Hard written in PLC:

defaultA = [xxx; yyy]

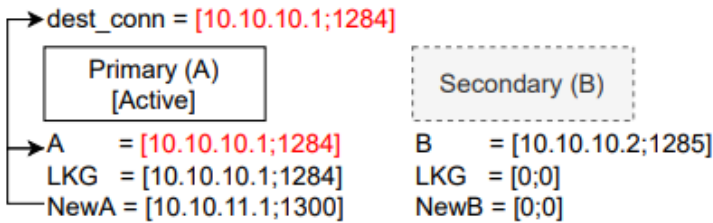
defaultB = [zzz; uuu]

dest_conn = [10.10.11.1;1300]	
Primary (A) [Active]	Secondary (B)
A = [10.10.11.1;1300]	B = [10.10.10.2;1285]
LKG = [10.10.11.1;1300]	LKG = [0;0]
NewA = [10.10.11.1;1300]	NewB = [0;0]

On a successful connection, the new connection details are loaded into LKG for storage and any flags indicating the change are cleared.

Step 4b: Connection Failed

Hard written in PLC:
defaultA = [xxx; yyy]
defaultB = [zzz; uuu]



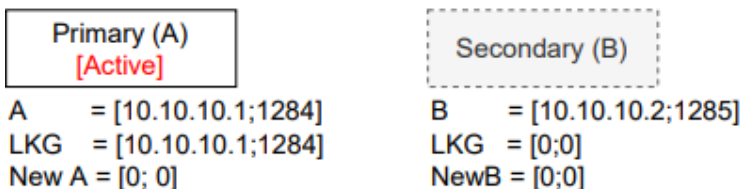
Should the connection fail consecutively for the set number of Retries, the system will then load back the previous LKG stored in A. **Please Note:** from here no flags will be cleared as the system as effectively returned to the state illustrated in Step 2. If you do not then clear CW13, 14 and 15 to zero or its previously set connection details, then on the next reset or connection fail it will attempt to connection on the new connection details.

8.3. Change Connection Details [Updating Secondary (B)]: Standalone

Step 1: System Overview

Hard written in PLC:
defaultA = [xxx; yyy]
defaultB = [zzz; uuu]

dest_conn = [10.10.10.1;1284]

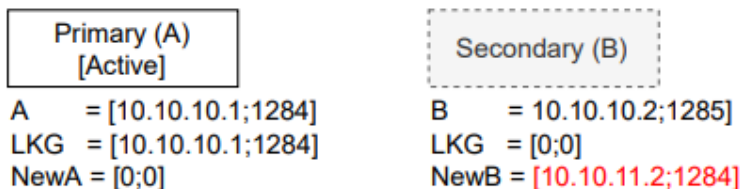


We are connected on A and we want to change the connection details of the Secondary (B).

Step 2: New Connection Details Received

Hard written in PLC:
defaultA = [xxx; yyy]
defaultB = [zzz; uuu]

dest_conn = [10.10.10.1;1284]



New connection details are received from the Agent Server (in this case since it is new B details, these are sent through on CW19, 20 and 21). Since we are in standalone, and the new connection details are on the Secondary connection (B), the system will **not** alter the configuration. **However**, from this point the operator can enable dual-port mode and at that point the changes will be made to the system. Review '**Change Connection Details [Applied to the alternate connection]: Dual-Port**'

8.4. Change Mode of Operation: Standalone -> Dual-Port

Step 1: System Overview

Hard written in PLC:

defaultA = [xxx; yyy]

defaultB = [zzz; uuu]

dest_conn = [10.10.10.1;1284]

Primary (A)
[Active]

A = [10.10.10.1;1284]

LKG = [10.10.10.1;1284]

NewA = [0;0]

Secondary (B)

B = [10.10.10.2;1285]

LKG = [0;0]

NewB = [0;0]

Connected on the Primary (A) and the changing to dual-port mode is done by writing any value greater than 0 on CW28.

Changing from Standalone to Dual-Port will cause the Secondary (B) to become available, allowing on a connection fail on A to connect on B and vice versa.

8.5. On First Ever Boot-Up: Dual-Port

Step 1: System Overview

Hard written in PLC:
defaultA = [10.10.10.1; 1284]
defaultB = [10.10.10.2; 1285]

dest_conn = [0;0]

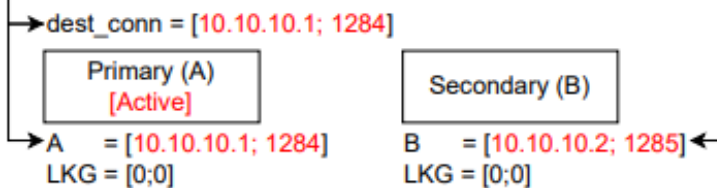


System in standalone does rely on a secondary connection (B), though its connection details can still be altered.

The PLC needs to be provided with default IP addresses that are inserted into the program before it is compiled and written to the device. The system will identify that when booting up in this state, since Last Known Good (LKG) for both connection details are zero, we must load the default details.

Step 2: Loading Connections

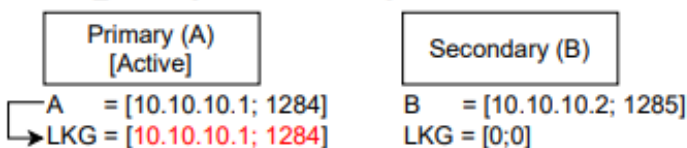
Hard written in PLC:
defaultA = [10.10.10.1; 1284]
defaultB = [10.10.10.2; 1285]



The system will load in the connection details for the primary connection (A) and the secondary connection (B), always attempting to connect on the primary first.

Step 3a: Connection Succeeded on Primary

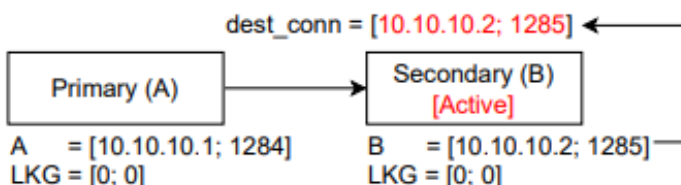
Hard written in PLC:
defaultA = [10.10.10.1; 1284]
defaultB = [10.10.10.2; 1285]
dest_conn = [10.10.10.1; 1284]



On a successful connection, the details of the Primary (A) will be stored in Last Known Good (LKG). The connection will remain on A. **Please Note:** To check the validity of the Secondary (B), ForceSwitch (AW0, B14) must be set to 1.

Step 3b: Connection Failed on Primary

Hard written in PLC:
defaultA = [10.10.10.1; 1284]
defaultB = [10.10.10.2; 1285]



On a failed connection, meaning in this scenario there was never a connection on A, the system will load in the Secondary connection details to the working details and connection is attempted. Should it connect on B, the details will be stored in the LKG related to B, otherwise it will reattempt on A.

8.6. General Connection Fail: Dual-Port

Step 1: System Overview

Hard written in PLC:

defaultA = [xxx; yyy]

defaultB = [zzz; uuu]

dest_conn = [10.10.10.1; 1284]

Primary (A)
[Active]

A = [10.10.10.1; 1284]

LKG = [10.10.10.1; 1284]

Secondary (B)

B = [10.10.10.2; 1285]

LKG = [0;0]

The system is connected on one of the two connections.

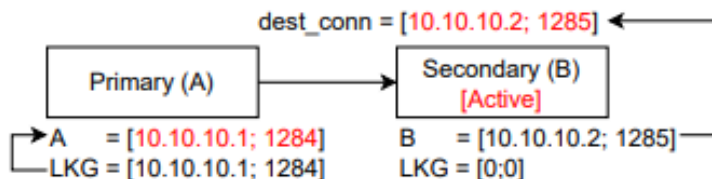
Please Note: The connection fail mechanism works the same way regardless if connected on the Primary (A) or Secondary (B).

Step 2: Connection Fail

Hard written in PLC:

defaultA = [xxx; yyy]

defaultB = [zzz; uuu]



When a connection fails (and cannot be re-established after the specified number of retries) the system changes to the alternative connection. During the switchover, the LKG of the connection that failed is loaded into connection detail slots. Any subsequent attempts on that connection will then be on a known good connection.

Step 3a: Connection Successful on the alternative connection

Hard written in PLC:

defaultA = [xxx; yyy]

defaultB = [zzz; uuu]

dest_conn = [10.10.10.2; 1285]

Primary (A)

A = [10.10.10.1; 1284]

LKG = [10.10.10.1; 1284]

Secondary (B)
[Active]

B = [10.10.10.2; 1285]

LKG = [10.10.10.2; 1285]

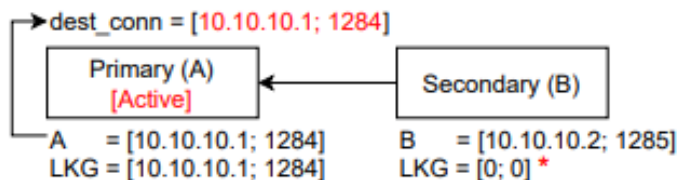
On a successful connection, the details of the connection will be stored in Last Known Good (LKG) as it relates to that connection. The connection will remain on that connection until a ForceSwitch (AW0, B14) is performed or that connection fails again.

Step 3b: Connection Failed on the alternative connection (without first successfully connecting)

Hard written in PLC:

defaultA = [xxx; yyy]

defaultB = [zzz; uuu]



On a failed connection (without any successful connection) the system will return to the connection details stored in the opposite connection.

*** Please Note:** If a successful connection was never made on one or more of the connections, LKG would be 0 and therefore the system will not load it to replace the failed connection details.

8.7. Change Connection Details [Maintaining the current connection]: Dual-Port

Step 1: System Overview

Hard written in PLC:

defaultA = [xxx; yyy]

defaultB = [zzz; uuu]

dest_conn = [10.10.10.1; 1284]

Primary (A) [Active]	Secondary (B)
A = [10.10.10.1; 1284]	B = [10.10.10.2; 1285]
LKG = [10.10.10.1; 1284]	LKG = [0; 0]
NewA = [0; 0]	NewB = [0; 0]

The connection is currently on A and a change is to be made on A.

Please Note: The example here applies to both the Primary and Secondary connections.

Step 2: New Connection Details Received

Hard written in PLC:

defaultA = [xxx; yyy]

defaultB = [zzz; uuu]

dest_conn = [10.10.10.1; 1284]

Primary (A) [Active]	Secondary (B)
A = [10.10.10.1; 1284]	B = [10.10.10.2; 1285]
LKG = [10.10.10.1; 1284]	LKG = [0; 0]
NewA = [10.10.11.1; 1300]	NewB = [0; 0]

New connection details are received from the Agent Server (in this case since it is new A details, these are set through CW13, 14 and 15).

Since we are in dual-port mode, a **Force Switch flag** will be raised (CW29, B3). To bring in the changes while **maintaining the current connection**, the **Reset_M2M** (AW0, B1) must be triggered.

Step 3: Connection details loaded

Hard written in PLC:

defaultA = [xxx; yyy]

defaultB = [zzz; uuu]

dest_conn = [10.10.11.1; 1300]	
Primary (A) [Active]	Secondary (B)
A = [10.10.11.1; 1300]	B = [10.10.10.2; 1285]
LKG = [10.10.10.1; 1284]	LKG = [0; 0]
NewA = [10.10.11.1; 1300]	NewB = [0; 0]

On a ResetM2M triggered by the operator, the new details are loaded into A (current connection) and the destination connection, where the connection will be attempted up to the number of retries set in the system to verify the details.

Step 4a: Connection Successful

Hard written in PLC:

defaultA = [xxx; yyy]

defaultB = [zzz; uuu]

dest_conn = [10.10.11.1; 1300]	
Primary (A) [Active]	Secondary (B)
A = [10.10.11.1; 1300]	B = [10.10.10.2; 1285]
LKG = [10.10.11.1; 1300]	LKG = [0; 0]
NewA = [10.10.11.1; 1300]	NewB = [0; 0]

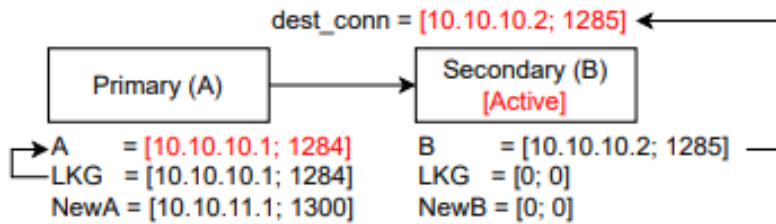
On a successful connection, the new connection details are loaded into LKG for storage and any flags indicating the change are cleared.

Step 4b: Connection Failed

Hard written in PLC:

defaultA = [xxx; yyy]

defaultB = [zzz; uuu]



Should the connection fail consecutively for the set number of Retries, the system will then load the connection details of the alternative connection (in this case, Secondary connection B) and restore the previous LKG stored in A.

Please Note: from here no flags will be cleared as the system has effectively not been changed. If CW13, 14 and 15 have not been manually set back to zero or to its previously set connection details that were accepted, then on the next ForceSwitch or connection fail it **will attempt to connect on the new connection details.**

8.8. Change Connection Details [Applied to the alternate connection]: Dual-Port

Step 1: System Overview

Hard written in PLC:

defaultA = [xxx; yyy]

defaultB = [zzz; uuu]

dest_conn = [10.10.10.1; 1284]

Primary (A) [Active]	Secondary (B)
A = [10.10.10.1; 1284]	B = [10.10.10.2; 1285]
LKG = [10.10.10.1; 1284]	LKG = [0; 0]
NewA = [0; 0]	NewB = [0; 0]

The connection is currently on A and a change is to be made on A.

Please Note: The example here applies to both the Primary and Secondary connections.

Step 2: New Connection Details Received

Hard written in PLC:

defaultA = [xxx; yyy]

defaultB = [zzz; uuu]

dest_conn = [10.10.10.1; 1284]

Primary (A) [Active]	Secondary (B)
A = [10.10.10.1; 1284]	B = [10.10.10.2; 1285]
LKG = [10.10.10.1; 1284]	LKG = [0; 0]
NewA = [0; 0]	NewB = [10.10.11.2; 1301]

New connection details are received from the Agent Server (in this case since it is new B details, these are set through CW19, 20 and 21).

Since we are in dual-port mode, a **Force Switch flag** will be raised (CW29, B3). To bring in the changes, the **ForceSwitch** (AW0, B14) must be triggered.

Step 3: Connection details loaded

Hard written in PLC:

defaultA = [xxx; yyy]

defaultB = [zzz; uuu]

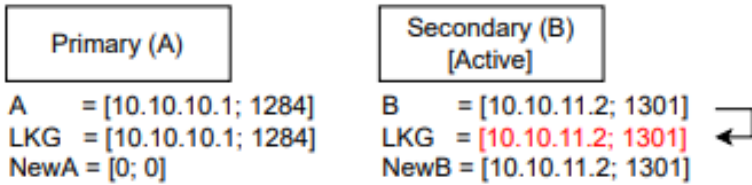
	dest_conn = [10.10.11.2; 1301]	
Primary (A)	Secondary (B) [Active]	
A = [10.10.10.1; 1284]	B = [10.10.11.2; 1301]	
LKG = [10.10.10.1; 1284]	LKG = [0; 0]	
NewA = [0; 0]	NewB = [10.10.11.2; 1301]	

On a ForceSwitch triggered by the operator, the new details are loaded into B and the destination connection, where the connection will be attempted up to the number of retries set in the system to verify the details.

Step 4a: Connection Successful

Hard written in PLC:
defaultA = [xxx; yyy]
defaultB = [zzz; uuu]

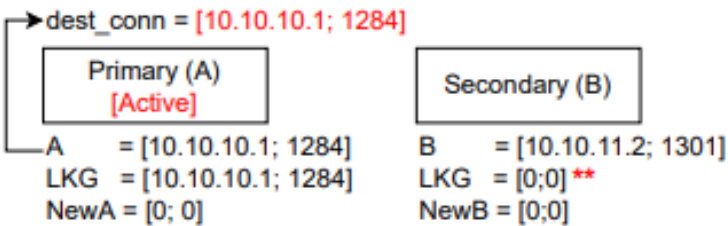
dest_conn = [10.10.11.2; 1301]



On a successful connection, the new connection details are loaded into LKG for storage and any flags indicating the change are cleared.

Step 4b: Connection Failed

Hard written in PLC:
defaultA = [xxx; yyy]
defaultB = [zzz; uuu]



Should the connection fail consecutively for the set number of Retries, the system will then load the connection details of the alternative connection (in this case, returning to Primary connection A). The LKG on the altered connection will be restored.

Please Note: from here no flags will be cleared as the system has effectively not been changed. If CW19, 20 and 21 have not been manually set back to zero or to its previously set connection details that were accepted, then on the next ForceSwitch or connection fail back on to B, it **will attempt to connect on the new connection details**.

** If a successful connection was never made on one or more of the connections, LKG would be 0 and therefore the system will not load it to replace the failed connection details.

8.9. Change Mode of Operation [While on Primary, A]: Dual-Port to Standalone

Step 1: System Overview (Connected on A)

Hard written in PLC:

defaultA = [xxx; yyy]

defaultB = [zzz; uuu]

dest_conn = [10.10.10.1;1284]

Primary (A)
[Active]

A = [10.10.10.1;1284]

LKG = [10.10.10.1;1284]

NewA = [0;0]

Secondary (B)

B = [10.10.10.2;1285]

LKG = [0;0]

NewB = [0;0]

Connected on the Primary (A) and the changing to standalone mode is done by writing 0 on CW28. Changing from Dual-Port to Standalone while connected on A will cause the Secondary (B) to become unavailable **without failing the connection.**

9. Revisions

9.1. Document Revision

Rev.	Date	Author	Comment	Approval
1.0	14-Dec-2022	J. Duncan	New Format	D. Jordaan
1.3	23-Jan-2023	K. Robinson	Adjusted formatting of images and sizing. Removed unnecessary references.	D. Jordaan
2.0				

9.2. DLL Revision

Rev.	Date	Changes
30	08-Sep-2021	FX5U_M2M_630.gx3 updated the M2M system function blocks to fix priority starving other functions. And fix to BigMoveR indexer registers being overwritten, Z2->Z1, Z3->Z4. Mit_M2M rev 30 (630) – Fixed rare buffer over read on “read data” but not always occurring.
30	23-Sep-2021	FX5U_M2M_630b.gx3 updated the M2M system function blocks to extend max DataMap count to be 500. (paired with Mit_M2M rev 30 (630))
34	17-Feb-2022	FX5U_M2M_634.gx3 History lockup fixes, Dual IP mechanism added (paired with Mit_M2M driver rev 33)
34	06-Apr-2022	QUDE_M2M_620b.gxw updated to include the standalone/dual-port configurations with the additional CONFIG WORDs and ACTION bit. (Paired with Mit_M2M driver rev 34). Mit_M2M.docx updated to include the describe the additions. Mit_M2M.docx updated to include the describe the additions.
35	26-Jun-2022	Added changes for the Shadowed and Parallel Scanning changes as part of mit_m2m.dll rev 35

9.3. Change Mode of Operation [While on Secondary, B]: Dual-Port to Standalone

Step 1: System Overview (Connected on B)

Hard written in PLC:

defaultA = [xxx; yyy]

defaultB = [zzz; uuu]

dest_conn = [10.10.10.2; 1285]

Primary (A)

A = [10.10.10.1; 1284]

LKG = [10.10.10.1; 1284]

NewA = [0; 0]

Secondary (B)
[Active]

B = [10.10.10.2; 1285]

LKG = [10.10.10.2; 1285]

NewB = [0; 0]

Connected on the Secondary (B) and the changing to standalone mode is done by writing 0 on CW28.

Step 2: Connection Moved to Primary

Hard written in PLC:

defaultA = [xxx; yyy]

defaultB = [zzz; uuu]

dest_conn = [10.10.10.2; 1285]

Primary (A)
[Active]

Secondary (B)

A = [10.10.10.2; 1285]

LKG = [10.10.10.2; 1285]

NewA = [0; 0]

B = [10.10.10.2; 1285]

LKG_B = [10.10.10.2; 1285]

NewB = [0; 0]

Changing from Dual-Port to Standalone while connected on B will cause the Secondary (B) to become the new Primary (A). All details are copied to the primary connection and the secondary connection becomes unavailable **without failing the connection.**